How design can help a coral reef

The importance of design and the design process in coral restoration.



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Abstract

Due to climate changes and human impact, the world's coral reefs are suffering. If nothing is changed, some of the species that rely on the reefs may soon become extinct. To help restore corals and possibly make them more resistant to environmental stress, scientists together with coral restoration workers are working on finding a solution to this problem. One approach is to build structures for corals to grow on. These structures serve two purposes, as they would also provide shelter for fish and invertebrates that used to live in the reef.

The structures are often placed on the seabed by divers, which means they need to be easy to handle underwater. While diving it is important to avoid exhaustion since that can quickly increase the risk of running out of air. It is therefore crucial that these structures are designed with a shape and material that is suited for this type of work. This thesis explores how designing these structures centred around the human interaction with them, can make it easier and safer for the restoration workers. This work has also explored the possibilities of integrating a design for sustainability perspective, where a more ecofriendly material was explored along with social and economic sustainability.

The project was carried out by an industrial design process based on a research through design approach, where methods such as observation, literature review, material experiments and prototyping were used to develop a concept suited for several stages of the coral restoration work.

The result is a modular structure concept adapted for on-site manufacture, transport, underwater assembly and coral planting. The shape of the structure is designed to make the handling easier and safer, this could make the restoration work more efficient in the future. Material experiment shows the potential of a possibly cheaper and more sustainable material option based on a cement-like mixture of lime putty, volcanic ash, sand and salt water. This might partially replace Portland cement in the future

This thesis contributes with a human-centred and material-oriented design perspective within the field of coral restoration.

Keywords: Industrial design, design process, human-centred design, research through design, coral reef restoration, coral reef structures, sustainable material

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Preface

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1 Introduction

Coral reefs are found in our oceans and are an important part of our marine ecosystem. WWF (2025) explains that corals are animals that live close to one an another and together they form an ecosystem known as coral reefs, where fish, invertebrates and other organisms can find shelter and food. Yuan, Lin, Pan and Yang (2024) says that coral reefs contribute to marine biodiversity, it provides food for many species including humans, acts as a natural wave breaker which protects the coastline and it supports tourism, and thereby contributing to local economics. It is therefore an important part of our planet and essential for a lot of species survival, including humans.

1.1 Background

Due to climate change and an increased human impact the world's coral reefs are suffering. Giallongo, Shashar and Rinkevich (2026) explain that if we don't put a stop to this it will lead to a loss of biodiversity, which will damage marine habitats and most likely lead to increased algae levels in the oceans. According to Strömberg, Rossi, Vonderscher and Larsson (2025) studies has shown that a damaged reef may have a limited ability to recover on its own. One way to help save the coral reefs and the species living there is through coral restoration.

Today people and companies around the world are working on restoring coral reefs. Personal observations and interviews indicate that in some parts of the world dynamite fishing (blast fishing) has been more common, which has led to not only dead fish but big gaps in the coral reefs. This type of illegal fishing leaves mainly sand behind, which creates a problem for corals to grow back. The coral larvae need a stable surface to grow on to develop into a coral polyp, especially in strong currents. This is why restoration teams often install some sort of structure in the sand, where the reef used to be. Some choose to plant corals directly on the structures and some place them close to the natural reef and wait for the corals to grow. While they wait the structures act as a hiding place and a home for the fish. Information presented in this section was obtained during volunteer work in Bali 2025, where structures were placed and studies about this topic were conducted.

There are some limitations in how the ecological theory is integrated in the work of coral restoration today Giallongo, Shashar and Rinkevich (2026) expresses. Some of the limitations they mention can be related to the design of the structures.

1.2 Problem

Coral reef restoration efforts has mostly focused on how to reproduce corals and how to increase the coral population Giallongo, Shashar and Rinkevich (2026) argues. Recent studies have begun to focus on the coral larvae and how corals can adapt and become more resilient. For example, Strömberg et al. (2025) have done a study on the coral larvae and how to attract them to grow in certain spots. Förster et al. (2024) have studied how corals can grow faster

and how the coral can be more resilient when exposed to volcanic ash. Despite this progress, limited research seems to be addressing the design role in this area, how can scientific findings be translated to practice? How can design make it easier for the people doing the restoration work, on remote islands or spots, to get the material on the spot, manufacture them at a convenient speed, transfer them to the spot and assemble them on the bottom of the ocean? Could this be done with sustainability as a focus?

So, the question is: Can you design structures for coral restoration to make it easier to place and assemble them in a material that is more sustainable?

1.3 Purpose

The purpose of this thesis is to explore how design and the design process can help coral restoration work. There are two design parts that plays an important role in this type of work, material and the structures shape. It is therefore necessary to look at both these areas and find new solutions to the experienced problems in each area.

First, this project aims to develop and find a more sustainable material to design these structures in, something that works better for the local communities and the environment. Many of the structures today are manufactured in concrete material and if you read Aduldejcharas (2025) article you see that concrete has a big negative impact on the environment. The author has done studies on a new material and implies that it could minimise the use of concrete in some areas. This indicates that new materials can be found and used, the focus here will be to minimise the use of concrete and find a more local material to replace it with.

Second, this thesis explores if restoration work might get more efficient if the human interaction with the structures is considered during the design process. This aligns with a human-centred design approach, which according to Norman (2013, s8), emphasises understanding of human interactions. In many coral reef restoration projects, divers are responsible for transporting, assembling and placing the structures. Since a diver’s working time is limited by their air supply, it would be safer and easier if the structures were simpler to handle under water.

By combining the practical aspects with scientific findings, the outcome may be even better in the future. This reflective approach relates to how Stappers and Giaccardi (2026) describes research through design.

1.4 Objectives

The project will give new knowledge to how local materials can be an option when producing structures and the role design can have in coral restoration projects.

1.5 Delimitations

This thesis and all the studies have been done in Sweden, which means that the climate isn't a tropical one and all the material in the studies has been collected close by. Volcanic ash has been stored from Gyali, in Greece, and the seashells are waste products from Nordic food industry and from natural deposits sourced from the waters around Norway, Denmark and the Faroe Islands. According to Förster et al. (2024) volcanic ash is not the same all over the world, it contains different metals and if there is the same one the concentration of those metals can differ. With that said this thesis aims to explore the idea of a new material that is locally sourced, but it is necessary to point out that it needs to be tested locally.

These studies focus on the impact design and material could have for the coral reef restoration work. It is exploring how design can help coral restoration work, not studying how the work helps the corals. The work that touches the area on what corals and coral larvae prefers is based on other scientific studies and should therefore be seen as a theory until properly tested.

1.6 Target group

Everyone interested in/ or working with coral restoration work. This thesis aims to give a broad perspective about multiple areas in this type of work, which can help optimising and make the work easier. It is important to specialise in certain areas, but it is as important to look at all the aspects of the work and make sure nothing is missed.

2 Theory

What can an industrial designer do for coral restoration? At World Design Organization (2026) you can read about industrial design as a profession. An industrial designer's job is to solve problems. The user and the user experience have a central role in this work, so it is essential to empathise and understand the user's needs to develop efficient solutions. According to the website, the work is not only about designing products. An industrial designer designs systems, services and experiences. The goal is to improve quality of life, which requires industrial designers to think about sustainability.

Design is a profession that is based on multiple practical knowledges, according to Schön (1983) visualisation is an important part of a designer's work. He therefore considers it to be relevant for the designer to know how to use techniques such as sketching, building prototypes and mock-ups. Exploring and reflecting is an essential part of the process and the practical techniques of the work often helps to develop one's ideas. He says that both reflection and creativity can be seen as artistic training, which makes art a central part of a designer's work.

2.1 The process

Design is often described as an iterative process. Wikberg Nilsson, Ericson and Törlind (2021, s30-35) describes iterative as something repeated, all the steps in the design process should therefore be done repeatedly. This is to help the designer explore different solutions and ensure that nothing is missed. There are a lot of different frameworks and approaches to help the designer with the process, one is the Double Diamond framework. Pyykkö, Suoheimo and Walter (2021) describes the Double Diamond as two diamonds with two steps in each diamond, discover and define in the first diamond and develop and deliver in the second one (see fig. 1).

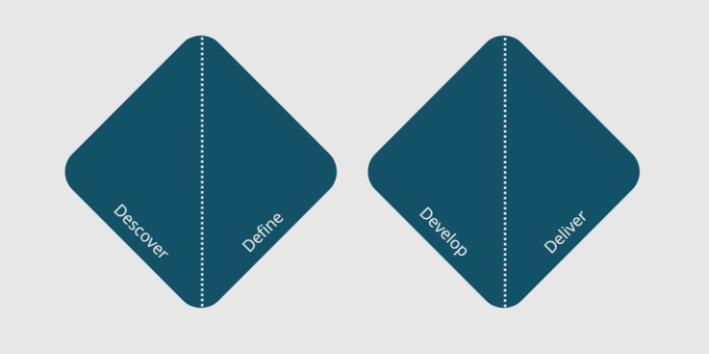


Fig 1. Double diamond framework.

According to Shen, Bosch, Pino and Gopalan (2024) the Double Diamond is not only for designers, it would be a good tool for scientists to use in their work as well. The article describes that when working with the double diamond it is important to use both divergent and convergent thinking. Wikberg Nilsson, Ericson and Törlind (2021) points out that it is also essential to focus on the process and to explore alternative options, not focus on a solution too fast.

2.2 Research through design

Researching is when someone is searching for answers through actions explains Frayling (1993). The field explored is known from the start, the researcher then gathers knowledge within the field to find answers. He says that a designer's practice shares common characteristics with the scientist research, they both search for new perspectives and information by using a practical and reflecting process.

Parts of the activities in a designers work is research, for example observations and interviews, says Stappers and Giaccardi (2026). This is how the designer gathers scientific knowledge to learn information about the situation they are designing for. They say that scientific theories can be somewhat abstract and the designer uses the design practice to fill the gaps and to find new information. Typical design activities can therefore generate foundational knowledge.

Something that brings the design and the science fields closer together is research through design says Stappers and Giaccardi (2026). They explain that a designer is trained in how to

understand and define valuable knowledge, in complex situations, in order to develop prototypes that address it. This is done in an iterative process. They mean that the designer can then contribute to valuable knowledge to someone else's research. By making prototypes the solution is visualised, which can both help explain the solution to others as well as helping the researcher generate new knowledge within the field. During this process the designer will realise what barriers that exist between the product and the user, this gives new knowledge and insights that can be shared with others. To share new knowledge with others, Frayling (1993) empathises the importance of documenting the full design process. This is how the designer can communicate and explain all the steps that led to the solution.

2.3 Human centred design

The human centred design approach is a method were the designer puts the human and the human needs in the centre of the design process. Norman (2013, s 8) explains that a lot of today's products and designs are causing frustration for the user, simply because they are not designed so the user can easily understand them. He proposes human centre design as a solution to minimise irritation caused by design. The design should communicate to the user how it is supposed to be used. To do this he implies that the designer needs to have a good understanding about the user of the product and what they need. According to Norman (2013, s8 & 221-235) this philosophy could be added to the double diamond design process to build this consideration and understanding. He explains that it is four iterative activities that needs to be done within human-centred design. These are: observation, idea generation, prototyping and testing. It doesn't matter what is designed, human centred design can be included in any process.

2.4 Design for sustainability

"Development that meets the needs of the present without compromising the ability of future generations to meet their own needs" is a common quotation by Brundtland (1987) cit. in Ceschin and Gaziulusoy (2016), when someone is referring to sustainable development. Ceschin and Gaziulusoy (2016) implies that even if the quotation focuses on the human needs, the work towards sustainable development has had its focus on the ecological sustainability. The reason for that could be because humans are depending on nature and its resources, in order to survive, but there are more aspects of sustainability to take into consideration. The article explains that it is important to work process based and to have a systematic approach to accomplish sustainability. They mention different aspects to consider in sustainable design, one thing is removing materials with a high environmental impact from the product. One other suggestion mentioned is to explore how the design can lower the energy used when a product is produced or disposed, by thinking about the whole life cycle of the product. Some critic is mentioned about this since this mostly focuses on the environmental part of sustainability. The article empathises the importance of implementing social and economic sustainability into the product in order to design for sustainability.

2.5 Material and design

Designers play an important part in achieving a sustainable design since more than 80% of the environmental impact a design will cause is decided in the design process says Italia, Papile, Santi and Del Curto (2023). One part of the design that can lower the environmental impact is the material. The article emphasises the importance of considering the use of resources carefully, as some resources are critical, this includes raw materials. It can be difficult to determine which material is more sustainable than the other, it depends on the context and how the product will be used explains Italia et al. (2023). The material will for example have impact on the production process, the suppliers/ availability, energy consumption and recycling. Therefore, it is important to have a holistic approach from the start of the process, says the article.

Portland cement is the material that causes the most carbon dioxide emissions in the construction industry says Aduldejcharas (2025). This is due to the production of the material. The article suggests that more environmentally friendly materials can be found and many are looking for good alternatives to this issue. According to Jativa, Ruales, and Etxeberria, (2021) cement is the second most used material in the world, only water is more used, and it has been used for a long time by us humans. They explain that a material called pozzolanic can be used to partially or totally replace Portland cement. A pozzolanic material is a material that, if mixed with lime (or an alkaline product) and water, gets similar properties to cement. If this is mixed with cement the article mentions several improvements to the finished product, such as a material that is more resistance to sulphates and an improved strength over time. There are both natural pozzolanic materials, for example volcanic ash, and artificial pozzolanic material, such as fly ash. Jativa, Ruales, and Etxeberria, (2021) make a compartment to the heating process, Portland cement is heated to 1350-1450 degrees Celsius to produce clinkers, which is the binder of the material. Pozzolanic materials need a much lower temperature, sometimes only 25 degrees Celsius. This lowers the carbon dioxide emitted significantly and the article mentions that pozzolanic materials are available around the world.

2.6 Coral reefs

Coral reefs are found in the oceans and are a valuable ecosystem for our planet. Yuan et al. (2024) explains that the reefs provide approximately 25% of the species living in the oceans with shelter or food. It is not only the marine animals that benefits from the reef, it also provides benefits to the humans that live close by the reef. The article mentions that millions of people all over the world get both food and income from the reefs, since they support fishery and serve as a tourist attraction. This makes the reefs an important part of the society.

Fish and invertebrates are an important part of the coral reef ecosystem says Loong et al. (2026) since they make the reef more resilient. The species living in the reef has different roles and are therefore important for the reef in different ways. Some examples giving in the article is that some species help control the algae growth, some help with the recycling of

nutrients and some help controlling the population of the species in the reef to keep balance within the system. They explain that all these species are an important part of the reefs ecosystem and that diversity helps the reef recover in case of any disturbances.

One of the biggest threats to coral reefs is increased temperatures in the water due to climate changes states Burdett et al. (2024). This is not the only threat according to Yuan et al. (2024), overfishing and industrial activity is major issues as well. This has led to a loss of corals, which means that the coral reefs area has decreased over the world. Ceccarelli et al. (2020) explains that measures have been implemented to reduce overfishing, improve the water quality and increase coastal progression. Unfortunately this is not enough, Strömberg et al. (2025) says that if a reef environment gets damaged there is a small chance for the reef to recover on its own. Strömberg et al. (2025) explains that areas that have been protected for many years have shown no visible sign of recovering on their own.

Along with measurements one approach is to try to restore the coral reefs. One restoration approach that Ceccarelli et al. (2020) mentions is a method where some sort of structure is placed on the seabed to create a stable area for corals to grow on. This can be done in spaces were a lot of rubble causes problems for corals to grow back, because of a lack of stable surfaces. The article explains rubble as loose coral fragments that is mostly due to blast fishing, storms and ship strikes. Ceccarelli et al. (2020) emphasises that there is a lot of knowledge gaps with this approach, since there is a lack of monitoring, but they can also see the potential in this since the areas caused by this are somewhat small. According to the article the structures wary in shapes and in sizes, materials mentioned is stone, concrete and steel. If there are coral larvae on the spot where the structures are installed, they can provide a stable ground for the larvae, create shelter for fish and in some cases even attract the larvae to the spot says Ceccarelli et al. (2020).

When it comes to materials in the marine environment Ceccarelli et al. (2020) emphasises the risk of an untested material in structures for coral restoration, since some materials can be beneficial to some corals but not others.

3 Method

3.1 Discover

In the beginning of the process it is important to discover and understand the problem, empathises with it Pyykkö, Suoheimo and Walter (2021) explains. A commonly used word in this phase is research, which Wikberg Nilsson, Ericson and Törlind (2021, s64-83) describes as when the designer explores existing studies, facts and knowledge within the field. They say that the importance for a designer here is not to understand how it is, more so how it should be. In this thesis the following methods were used in this phase:

- Observations
- Survey

- Literature review
- Interviews

3.1.1 Observations

To get a deeper understanding of the problem it might be relevant to experience it or to observe it. According to Wikberg Nilsson, Ericson and Törlind (2021, s64-83) this helps the designer to identify user needs in a context. It is important to see, listen, ask questions and experience what the user does. Norman (2013, s8) suggests that observation is a good method for incorporating human centred design into the design, since this can help to give knowledge to problems that the user hasn't even thought of.

3.1.2 Survey

In Blomkvist and Hallin (2014, s82-90) book the method survey is suggested when it is necessary to obtain general and/ or quantitative information. For a designer, especially an industrial designer who usually tries to find a solution to a problem that many people experience, this could be a good method to use for collecting data and statistics.

3.1.3 Literature review

A thesis is a scientific work, it is therefore necessary to investigate what is already done within the explored field describes Blomkvist and Hallin (2009). This is commonly done by a literature search, which includes but is not limited to books, articles and papers. They explain that it can be hard to find relevant literature, usually it is a good approach to read more literature then needed to get a broad perspective.

3.1.4 Interviews

The interview method is a good way to collect qualitative information about a subject explains Blomkvist and Hallin (2009). To conduct interviews in the beginning of a thesis work can help to understand the problem and narrow down the work they continue. Different ways to conduct interviews, depending on what information needs to be collected, is described in the book. The most common one they describe as a semi structured one where the interviewer has prepared questions before the interview, but adapts the questions during the interview, to fit the person being interviewed.

3.2 Define

In this phase you define the problem and work on narrowing down the information you got in the discover phase Shen et al. (2024) describes. They explain that this can help challenge the assumptions ones can have starting a new project or find the actual core to the problem.

This phase is thus built on the knowledge and research gathered in the previous phase, this means that different designers could understand and prioritise the problems differently explains Dorst (2003). This personal interpretation can have an important role in a design process where the problem is subtle.

To define the information from the discover phase these methods were used in this thesis:

- Function analysis
- Knowledge map
- System map
- Activity analysis

3.2.1 Function analysis

A function analysis will help the designer to understand the reason for the solution Wikberg Nilsson, Ericson and Törlind (2021, s75) explains. It is common to define the need and the function with a verb and a noun, the book gives the example provide transport if the product is a bike. This doesn't provide a solution, but it gives an understanding about the solution.

3.2.2 Mindmap

According to Wikberg Nilsson, Ericson and Törlind (2021, s49) this method is used to organise collected research, ideas and other aspects of the project. In a mindmap information is written down about the subject and then grouped based on similarities. By using this method, the designer can get a clearer overview of the project and identify key components.

3.2.3 System map

A system map can visualise and help to analyse complex systems, by identifying key components and their relationships according to IDEO U (2024). This will help the designer to understand how the solution will affect the system. When making a system map, the stakeholder within the system is written down on a paper, lines or arrows are drawn between parts that are connected to each other.

3.2.4 Activity analysis

With this method the designer is writing or sketching down every step of an activity explains Österlin (2016, s43). Unlike the system map, which focuses on the relationships within the system, this method focuses on the actions that someone does within a system or a process.

3.3 Develop

In this part of the process it is time to create ideas and solutions, divergent thinking. According to Wikberg Nilsson, Ericson and Törlind (2021, s116-119) it is important to think in new ways about the ideas and to be creative. Österlin (2016, s63-64) says that creativity is when a problem is solved in new ways. He implies that creativity can be helped with openness and motivation, it is hard to be creative with too many demands. It is also necessary to know what the goal is and to question what already exists and is established.

To develop new ideas and validate them, these methods were used during this project:

- Brainstorming
- Sketching
- Analogies
- Prototyping
- PNI

3.3.1 Brainstorming

This method can be used in many different ways but the goal with a brainstorming session is to get a lot of different ideas, according to Wikberg Nilsson, Ericson and Törlind (2021, s125-137). For example, you could choose a theme and try to create ideas within this theme. The most important thing is to feel safe and to not criticise any ideas that comes up during a brainstorming session, a safe space helps to generate new ideas.

3.3.2 Sketching

Sketching is a designer tool that often is used during the process. It is a fast way to try out ideas in the form of a drawn image says Österlin (2016, s93). If combined with a different method during the develop phase, a new idea can be both born and visualised.

3.3.3 Analogies

Analogies are a method to help the designer to think in new ways through metaphors. Wikberg Nilsson, Ericson and Törlind (2021, s127 & 153) describes that a concept is written down on a piece of paper, then the designer fills the paper with ideas on how a problem can be solved with the help of the concept. They explain that it is important not to think too rationally when using this method, since it is used to get new perspectives and to help with creativity.

3.3.4 Prototyping

Prototyping is when you try out your idea in a physical model explains Wikberg Nilsson, Ericson and Törlind (2021, s 170-175). A prototype helps explore the ideas, develop them and in the end validate them. A variety of materials can be used, one example they mention is clay. An easy material to work with which can help communicate the ideas or support new ideas.

3.3.5 PNI

Blomkvist and Hallin (2014, s49-52) says that it is important to validate the ideas in a thesis work. They explain that validation in this context means that the student is studying the right thing. In a design work it is also important to validate the solutions against the problem. Wikberg Nilsson, Ericson and Törlind (2021, s165) explains PNI, one of the many ways to do this. All the ideas are written down and then the designer writes down something positive, negative and interesting (or according to the book neutral) about the idea. This helps validate the solutions and compare them to each other, convergent thinking.

3.4 Deliver

This is the final step of the Double Diamond framework, it is now time to finalise the solution. For a designer it is important to present the solution in a way that it is understandable for others. Wikberg Nilsson, Ericson and Törlind (2021, s198-213) talks about the importance of motivating the design. It is usually not a coincidence why a design is shaped in a certain way, this needs to be explained when presented. They suggest that one efficient way to do this is through visual communication. It is easier to understand and remember the design if demonstrated in words and seen, they explain.

The methods used for communicating and visualise the design concept was:

- Material samples
- Presentation models
- Scenario

3.4.1 Material samples

According to Österlin (2016, s160) there are three aspects to investigate when choosing a material for the design. Which characteristics is required from the material, does it need to be light or sustainable for example, is it possible to produce the product in the material and is the material available?

In research through design it is helpful to make prototypes to both gain new knowledge and to visualise the design for others explains Stappers and Giaccardi (2026). Material samples can therefore be developed for the same purpose.

3.4.2 Models

This is an efficient way to visualise the finished result if the result is a physical product. A model can be created in many ways and materials. Österlin (2016, s91-99) implies that it can be hard to combine everything, such as material, shape and function, into one model. As a solution to this he proposes that one can create two models, one function model that focuses on the design function and one form model that focuses on the design shape.

3.4.3 Scenario

A scenario is a way to tell a story about how the user interacts with the design in a certain situation, Wikberg Nilsson, Ericson and Törlind (2021 s155) state. The designer can use this method both in the early stages of the process, to describe an existing situation, and later in the process, to help visualise how the new solution is intended to be used.

4 Execution

4.1 Planning

This thesis work started with a schedule and a logbook. The thesis was conducted within a research through design approach but the iterative process was structured by using the Double Diamond model.

4.1.1 Schedule

An initial schedule was made to make sure the project could be done on time. As earlier explained, design is an iterative process which often means that it is hard to set a strict schedule, as a designer it is important to be flexible. With that in mind the schedule was created with possibility to adapt certain parts but to make sure all deadlines were met (see the finished schedule, after adapting week by week, appendix A).

4.1.2 Logbook

Every day a logbook was kept. This is an important part of the process, especially in this iterative research through design process. Sometimes you need to take a few steps back and look at the problem from different angles. A logbook makes it easier to remember and to make sure nothing is missed. It is as important to go back and read it, as it is to write it. During this process it was the reflecting part of the process that helped get new insights.

4.2 Discover

4.2.1 Observation and field work

The idea of this thesis started during volunteer work in Bali. The work was based in northern Bali where a coral reef restoring team was placing structures for corals to grow on. The structures were made by locals and then dropped into the ocean by boat, where divers carried them (see fig. 2) and placed them close to one another for support. The currents made the work hard, the structures need to be relatively heavy and have a shape that can be pushed down in the sand, to improve stabilisation. The weight and shape of the structures made them difficult to carry, so when there were a lot of currents the work was especially difficult. This requires a skilled diver in physically good shape to not run out of air, which can be dangerous. On multiples occasions some of the divers in the team needed to abort the work and ascend earlier than planned, because they were running low on air. The structures had a rebar that were somewhat visible, this can also create a safety issue since they are sharp. It happened that people in my team got scratched by the rebars.

The structures were carried two by two and when they were placed by the project manager, I observed him trying to push them into the sand and secure them next to one another. Sometimes when visiting the same spot again, I noticed some of the structures had tipped over because of the strong currents. (See fig. 3 for placed structures).

Observing and performing this work was essential for me to experience the problem. During this volunteer work I talked to the people managing the project and other volunteer workers. They whished for the structures to be easier to carry close to the body, lighter individual structures that can be carried one by one and easier to place the structures in the sand (dig them down). I also learned that it was hard for the workers that needed to load and unload the boat with the structures, because of the shape and the weight. They informed me that it would be easier if the structures had a natural spot where a rope could be attached, that would make it easier to lower the structures down to the seabed. This helped me realise that a human centred design approach would probably help the handling of the structure and as Norman (2013) puts it, minimise irritation caused by the structures design.



Fig. 2: Two divers carrying a structure, photo from a video by: Ketut de Sujana.

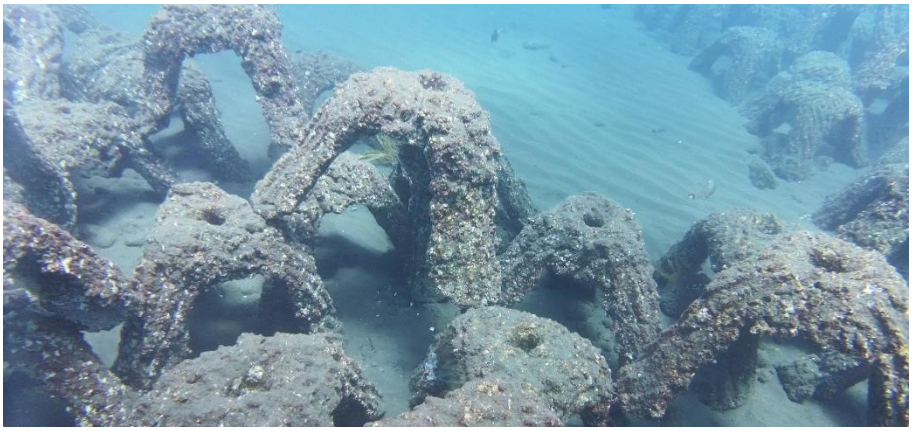


Fig. 3: Placed structures.

4.2.1.1 Different structures

As a diver, I have visited different locations and countries where coral reefs restoration projects were carried out. Variations in the shapes of the structures used were observed across these sites. Fig. 4 presents sketches of structures observed in multiple locations. At the beginning of this project online research was conducted to identify companies involved in coral restoration work, in order to get in contact with them for interviews. During this process, many of the companies were found to presented pictures with similar structural forms on their website. These three shapes may therefore represent some of the more commonly used shapes.

The box-shaped structure on the left in fig. 4 has been observed in both metal and concrete materials. The middle structure, resembling a cage, has been observed in a metal rebar, while the structure on the right has been observed both metal and plastic materials.

A reflection on these observations is that plastic material is generally lighter, which may be easy to transport, however I question their suitability in a marine environment. Structures

made of metal and concrete are heavier, which may complicate transports but could also improve the stability on the seabed. The box-shape structure on the left may also be more difficult to push down in the sand, compared to the other two forms.

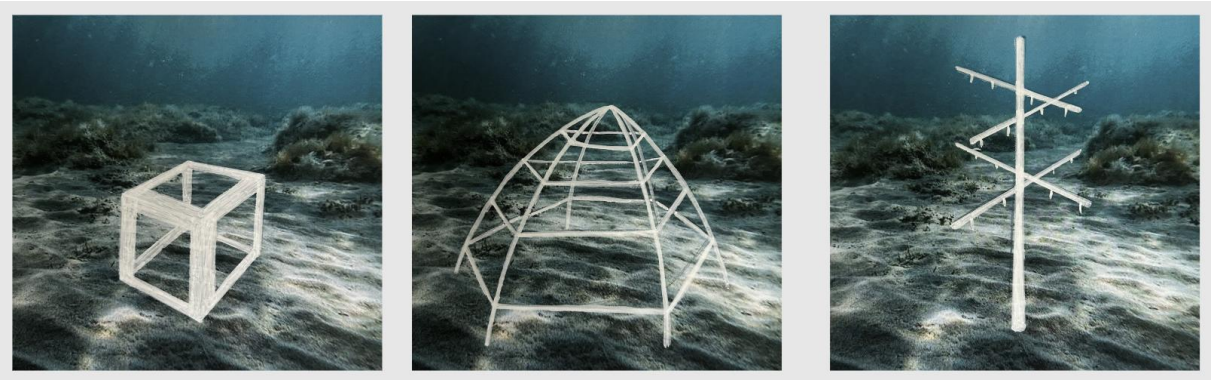


Fig. 4: Sketches of observed structures. Background photo by David Boca (Unsplash).

4.2.2 Survey and information through email

To get a broader understanding of the problem and what people working with coral restoration needs to make the work easier, 80 different places were contacted through email. The contacted were a variety of diving places, scientists, volunteer workers and engineers within the field. Along with the email a survey was sent out to all these places/ people (see appendix B for full survey). I learned that many people work with steel bar structures and/ or concrete ones, which often requires a rebar. These materials are not the most environmentally friendly and according to Strömberg et al. (2025) study it seems like coral larvae may not prefer a rusty surface, which cannot be avoided if any metal material is used. Although the responses to my survey were limited, 18 responses, they provided insights and information on how to prioritise the challenges. The survey and information through emails suggested that the material of the structure is one of the most important aspects of the design, together with stability on the seafloor, textured surface and a shape with nooks and holes for fish to hide, (see table 1). With this information I think that material availability could be a reason why many choose the materials they do. It is also important to understand that weight, modularity and stability on the seafloor is crucial for this type of work, which a steel bar structure can contribute to.

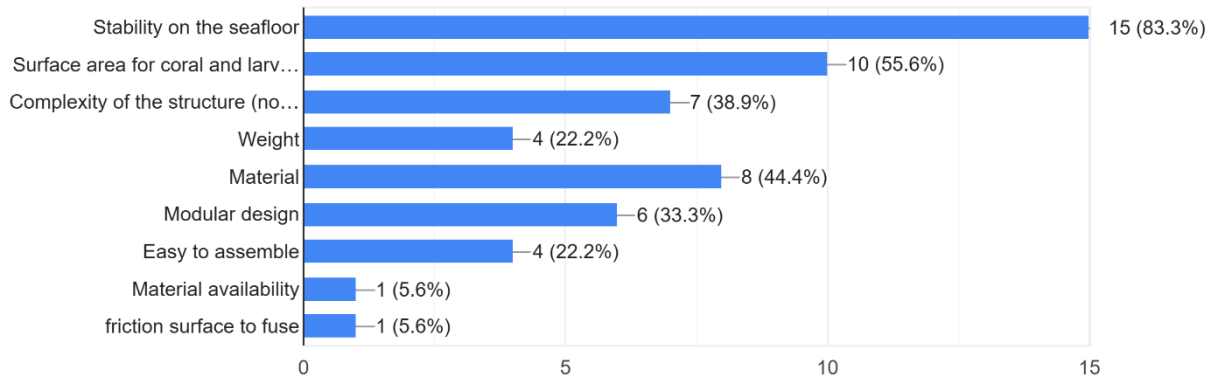


Table 1: Survey answers to the question: what aspects of structure design are the most critical in your experience? (Select up to three, if you want to write a comment choose option “other” as well). I got 18 answers to this question.

I learned, by the email answers I got, that people are trying to find more sustainable materials to build the structures in (see appendix C for questions sent in emails). This information made me reflect on how the new material should preferably be a local material that holds together for years and years in sea water. It needs to be easy to manufacture the structures in, provide a structure to the surface and a natural material for the ocean. Would it be possible to find marine-based material or something on the beach? Would it then be possible to manufacture the structures right at the beach?

4.2.3 Literature review

To get greater knowledge for the project I read multiple studies on design, material and coral restoration. Some of the articles were found on Discovery, the University of Gävle’s search engine, and some of the articles was recommended by scientists I got in contact with during the project. Search words on Discovery used for finding literature were: “coral restoration”, “coral restoration AND “structure”, “tropical reef environment”, “waste” AND “material design”, “coral” AND “volcano ash”, “research through design”. The rest of the literature in this thesis was included in the design courses, taken at the industrial design program at University of Gävle.

This thesis is exploring how design can help coral restoration work, not studying the corals need. With that said the structure’s main function is to work as a home for fish and corals, so it is crucial to look at aspects that are important for those creatures as well. Therefore, it was necessary to gain good knowledge and understanding about other scientist’s work that is based on what is essential for the corals. The literature has been used as a foundation for what design, material and shape, the corals in theory would prefer. This helped me realise that corals are animals that require an environment that doesn’t change too abruptly, to not cause stress on the animal. For example, quick temperature changes in the ocean would be stressful according to Burdett et al. (2024). After reading these articles I tried to put this information into what it would mean for the design of the structures.

Reading Giallongo, Shashar and Rinkevich (2026) article it is explained that the planning of the restoration work is important. They recommend restoration workers to look at how different species help each other and how and where corals should be planted for the best results. They mean that if the corals are planted in a way they can help and rely on each other, in the environment the corals need, the survival rate would be higher, and it would lessen the need of human intervention.

Their proposals on how to improve the restoration work are good in theory, but some might be difficult to implement in practice. It would require a broad knowledge, not only about one coral species but all of them. It would also require some knowledge about currents and the specific spot where the corals intend to be planted. This planning could cost a lot of time and funding, to get that type of expert knowledge. According to my research, funding is

something that a lot of restoration places struggle with already. To be able to implement the changes they suggest it must be easy for the people doing the restoration work, otherwise I think it is likely that it will stay as a theory. I believe that their suggestions are valuable and should be explored but it also needs to be practical to implement. This might be something that design can help with. For example: one of their proposals is to plant the corals closer to each other, of course it is necessary to look at what species should be planted at that specific spot and close to one another before you plant them too. But if we focus on what the design of the structure can provide is the possibility to make it easier to plant multiple corals on the same structure, for closeness and to make sure the structures are placed close to one another. The closeness could potentially save time for the diver, and therefore maybe even save money, because you can theoretically plant more corals on fewer dives.

To have growing corals it is important to look at the coral larvae and what they need to settle and develop into a coral polyp. Reading Strömberg et al. (2025) article gives good knowledge about this and their studies suggested that a structured surface may influence the coral larvae settlement. They tested different samples of material, and one was made with the help of a 3D-printer. According to the study the larvae seemed to like the surface of the 3D-printed one. They also seemed to prefer the concrete made from metallurgic slag, which has a higher content of magnesium. The study in this article is done on larvae from the coral *Lophelia pertusa.* This is a deep-living cold-water coral, so the results in this study may not correspond to all corals but I believe that the result is still interesting and worth investigating further.

Looking into another article about *Lophelia pertusa,* it is clear that more people are looking for a more sustainable material to design structures for coral restoration. Strömberg and Larsson (2025) explains in this article that the material of the structures needs to last in seawater for roughly one century, and here they say that concrete is known to be a good material for this. They mention Roman concrete as the most long-lived concrete known today and how they used a mixture with cement and volcanic ash. This mixture seems to make the material even harder than Portland cement while exposed to seawater. This is something that I find interesting since I know that volcanic ash contains magnesium, among other minerals. After learning this I ask myself, could volcanic ash potentially attract and influence the coral larvae attachment to the structure because of the amount of magnesium the ash contains? Would it also make the material more resilient to salt water?

Researching more about mixing cement and volcanic ash to produce the structure in, I found the article from Förster et al. (2024). They have done some promising studies on what happens to corals if they get exposed to a controlled amount of volcanic ash. According to the studies, that was made in controlled conditions, the corals seem to handle stress better and get an increased photosynthesis thanks to some of the metals in the ash, which can help reduce stress caused by increased temperatures in the ocean. It is important to point out that this was tested in a controlled environment with a controlled amount of volcanic ash. However, as a designer this is interesting, this suggests a potential material for the design. Especially since volcanoes seem to be close to the location of the coral reefs, according to the article. So now I am thinking, what if you could put an amount of volcanic ash in the material, could that

potentially help the corals from the stress caused by higher temperatures in the ocean and lower the risk for coral bleaching?

As earlier mentioned an article from Aduldejcharas (2025) was also read. In this article, a new type of waste material that could replace cement, in some cases, was explored. With shells from an invasive species of shellfish common on rice fields in South America and Asia. A powder was made of collected shells and then mixed with cement, water and sandy loam, this mixture was then pressed to compressed earth blocks using a machine. This material would not work for marine environments or at a remote island, according to the tests done on water absorption and the fact that it needs to be pressed by a machine. But it is an interesting approach to the problem on how to minimise the use of Portland cement and how to make a more sustainable material. Shells are common in the sea, and these shells should be more water resistant than the ones used in this study. Could it be possible to produce a binder of seashell waste to mix the volcanic ash with?

4.2.4 Interviews

To learn more about volcanic ash as a material, a company working with this material in Sweden was contacted. They provided information about their product, that comes from a volcano in Greece (see fig. 5). The stone is milled to a fine powder that acts as a pozzolanic binder (a type of ‘glue’ that will make the material stronger over time) when mixed with cement. Compared to Portland concrete this product has a significantly lower climate footprint. Producing 1 ton of Portland cement releases 800-900 kilo carbon dioxide while volcanic ash only releases 50-70 kilos. This means that if you replace a percentage of cement with this volcanic ash (usually between 10-30%) you could lower the negative impact the finished concrete material has had on the environment. According to the company the material contains no harmful substances to their knowledge, and it is also used for planting. They have for instance noticed biological growth (such as small plants and moss) on the material after having kept it outside, which could indicate that plants accept the material.

It is possible that this material is more water resistant and could have better corrosion resistance than using only Portland cement. The most crucial part is to mix the product with the right amount of water, if mixed with too much water it will take time for the material to settle and it is possible that the result will be a weaker material. The material strength over time is comparable to Portland cement, because of the pozzolanic reaction, but in the early stages the strength is less than Portland cement. It is therefore important to let the material be cured for 28 days.

The product has not been tested in salt water, yet, but it leads to a discussion about how they made and mixed concrete during the roman period and how they used volcanic ash, slaked lime (lime that is produced by burning calcium-rich materials in high temperatures and then reacting it with water) and water. Some of the harbours and piers made from this Roman concrete are still standing and the concrete was curing in the water, so in theory it should be possible for the material to sustain seawater. According to studies the company has done on the material, concrete with their product has shown a higher resistance against sulphates if

mixed with 20-50% of this volcanic material. Sulphates are found in seawater so this would hopefully eliminate cracks in the structure caused by sulphates.

This information appears promising for my project, and it suggests that my material theory could have advantages compared to Portland cement in seawater environments.

I also have an idea about mixing the cement with seawater that I discuss with the company. They seem to think it would be possible, but they haven't tried that yet. My thought is that if this would be possible, it would reduce the need to clean the seawater, use the limited rainwater or buy bottled water. That would probably make it easier and cheaper to manufacture the structures and make them even more sustainable.

To get a broader knowledge about Portland cement, how it is used and mixed two people working with concrete in the building business were contacted. They provided information on curing time, reinforcement of the material, how to mix it and what to think about when doing samples (see appendix D for all the questions asked in the interviews).

After the first interviews the company and these two construction workers were contacted on multiple times during the process. They provided insights and gave feedback on ideas and how to manufacture the structures.

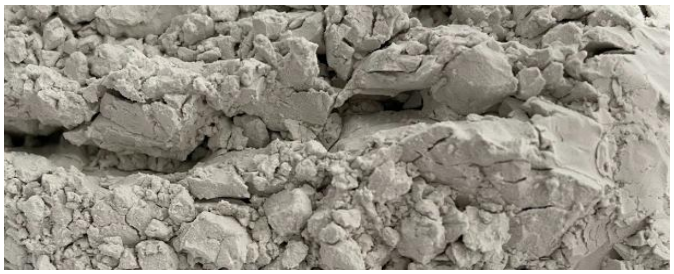


Fig. 5: Volcanic ash

Summarise discover phase:

- Observations indicated that human centred design needs to be incorporated into the process of designing the structures.
- It is important to look at the full process and all the people that will handle the structure.
- The structure needs a shape that is easy to load and unload from the boat, preferably have a spot where a rope can be attached, it needs to be lighter and/or easier to carry close to the body and it needs to be easy to "dig" the structure into the sand during the placing phase.
- A light weighted structure will be easier to transport but a heavier will be more stable.
- There is a need for a more sustainable material that does not rust.
- A local material could make it easier to manufacture the structures.
- The structure needs holes and a texturized surface for fish and coral larvae.

- Stability on the seafloor is important and the structures needs to be placed close to one an other.
- The structure needs to make it easier to plant multiple corals on it.
- The material of the structure could potentially attract coral larvae.
- It could be possible to produce the structures in a material that is more sustainable and local.
- The same material could potentially help the corals become more resilient to stress.
- Interviews helped understand the process of mixing and working with concrete.

4.3 Define

4.3.1 Function analysis

In the discover phase research was conducted in a lot of different areas. When you get that much information about different aspects of the project it can sometimes feel a bit overwhelming and it is hard to know where to start and what to focus on. This happened to me during this project. The design work is iterative, so working back and forth between the phases is necessary, but at one point I found it hard to develop any ideas and to find the creativity. I remembered a method I had used previously that helped prioritise the functions that the finished design needs. This helps sort the information and focus on the problems in order, instead of focusing on them all at once. It is also helpful to not forget what the main focus should be and to find new ideas for the solution. I believe that this is what Österlin (2016) meant when describing how knowing the goal will support the creativity.

Fig. 6 shows the main function, partial functions and support functions for the design with the most important one at the top. With one noun and a verb a function has been specified. In the next step of the Double Diamond, solutions for these functions can be explored.

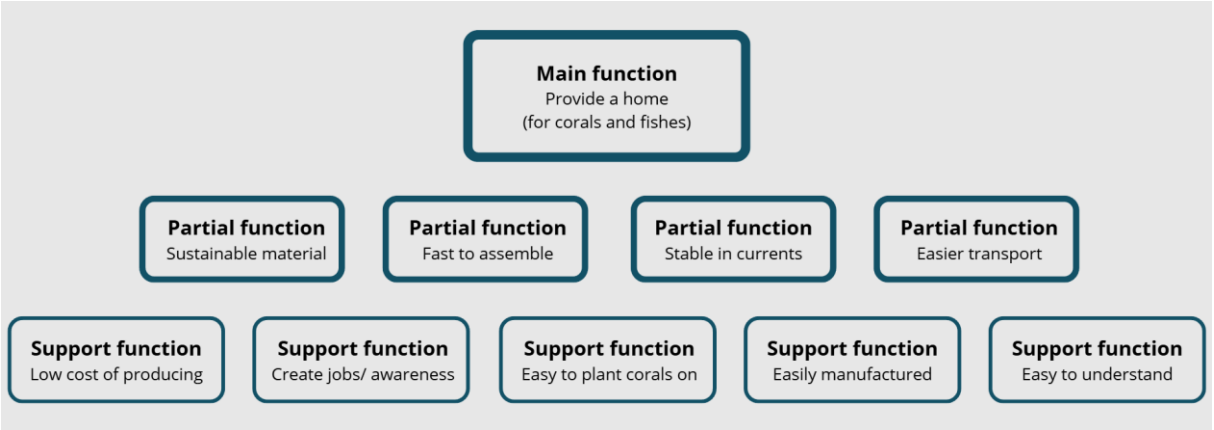


Fig. 6: Function analysis shows what functions are important for the design solution.

4.3.2 Mindmap

Information about the project was written down on sticky notes and then categorised into four categories. The categories were local materials, manufacture and transport, assemble and stability and fish and corals, which represents the main function and all the partial functions in the function analysis (fig. 6). The information was sorted by colours (see fig. 7).

While analysing these categories opportunities, risks, insights and design demands were written down on separate papers and categorised by the same colours to show which group they were related to (see fig.8). For example, in fig. 7 the yellow colour shows information related to manufacture and transport. In fig. 8, the notes related to that topic is therefore written on a yellow sticky note. This helped to organise the information gathered during the research and to find key aspects.

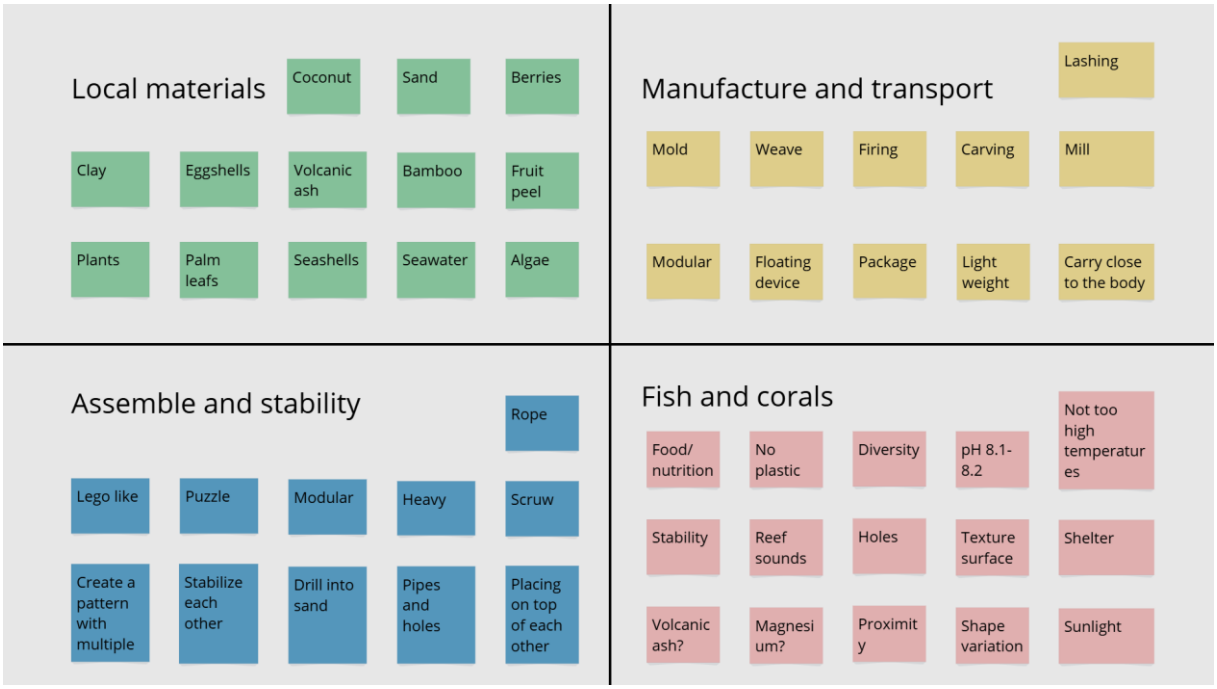


Fig. 7: Mindmap, information is categorised according to the main function and partial function from the function analysis.

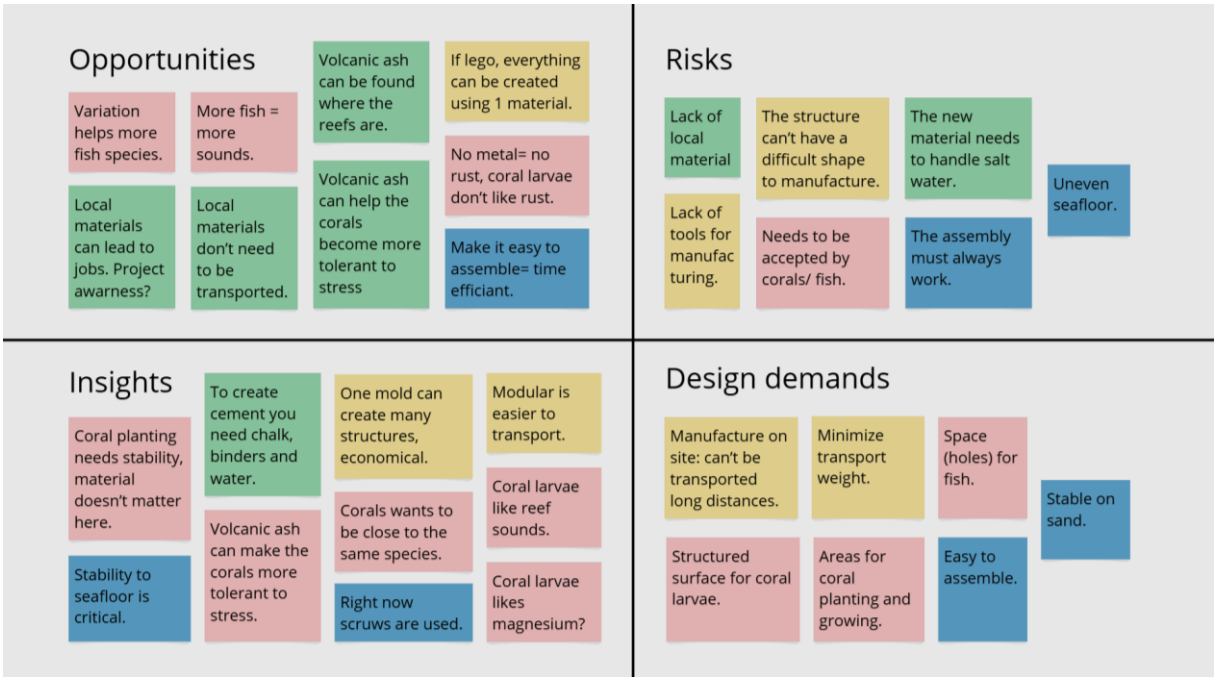


Fig. 8: Opportunities, risks, insights and design demands written down on coloured sticky notes. The colours represent the categories from the mindmap.

4.3.3 System map

As earlier said, Ceccarelli et al. (2020) emphasises the risk of an untested material in structures for coral restoration, because it could be beneficial to some species and not to others. For this reason, I wanted to do a system map to get a greater understanding of how everything is connected. This would eliminate the risk of developing a solution that could have negative effect on marine life.

In fig. 9 the middle explains what the system map is about, structures for coral restoration. The most important design factors, according to my research, are the circle closest to the middle. Next are the biological and environmental factors, this includes both what the structure must withstand, such as salt water, and the effect the structure, once placed, will have on the species living there. The outer circle represents the societal and logistical factors. After writing down the different aspects in the correct circle I made a line between the ones I could see were connected. Looking at the finished map, I realised that material was one central thing in the system and it affects multiple aspects in all three circles.

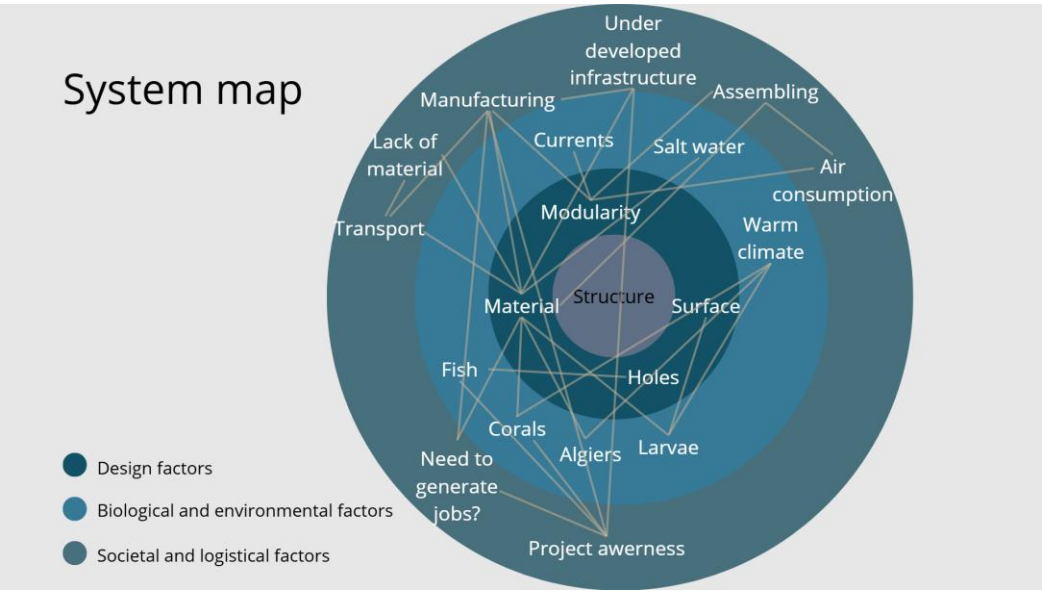


Fig. 9: System map on how a structure affects the parts in a coral reef system.

4.3.4 Activity analysis

This method was used to understand the structures stages and what the work looks like, from building the structure to maintenance (see fig. 10). Normally every step would be written down in an activity analysis but here the focus has been on the steps that have shown to be the most problematic ones from a designer's point of view.

Step 1- the structures are built, this requires material and tools. Most often they are built close to the beach and by hand, not industrially produced.

Step 2- The finished structure is loaded onto a boat, often a smaller fishing boat, and transported to the drop-off spot. Here the structures are dropped down or sometimes given to the divers, which means that the divers are then transported with the structures.

Step 3- Divers and their equipment are transported to the spot. They dive down to the dropped structures or go down with the structures.

Step 4- Divers carry the structures to the spot where they should be placed.

Step 5- The structures are placed close to one another and somehow secured to the seafloor. They are pushed or dug down in the sand or attached to a rebar on the seafloor.

Step 6- Coral fragments are planted on the structure. This is something that requires analysis and maintenance. It is common to clean the structure and to monitor and analyse the coral to make sure it attaches to the structure.

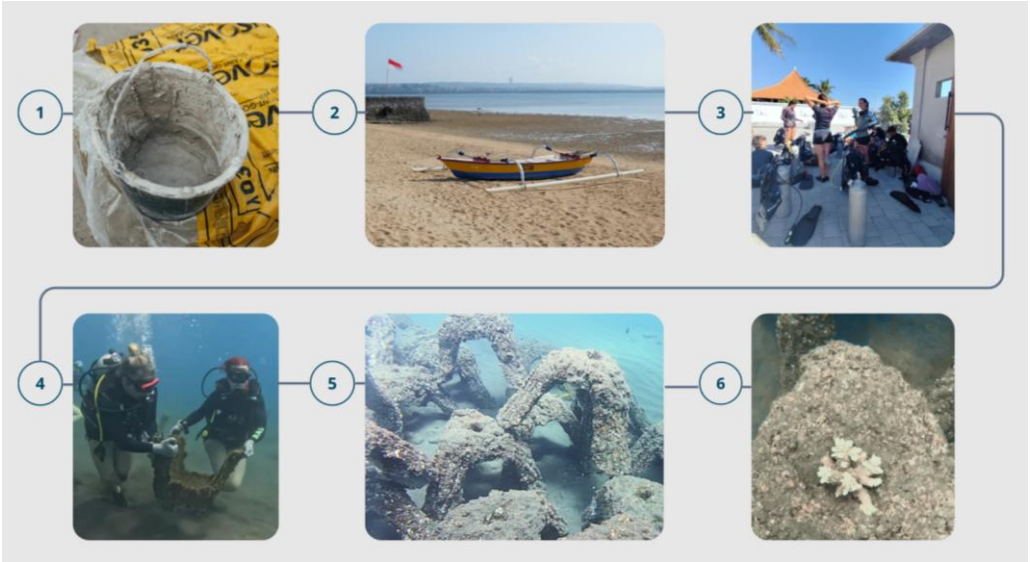


Fig. 10: Activity analysis of the more critical parts in coral restoration work. Photo 1 by Benjamin Jopen (Unsplash). Photo 2 by Safira Nur Fajrina Putri (Unsplash). Photo 4 by Ketut de Sujana.

To define the information I gathered about the material in the discover phase, I made one more activity analysis (see fig. 11). This time the focus was on how cement is produced and transported before it reaches the location where the structures are built.

Step 1- Limestone is quarried.

Step 2- The stone is transported to the factory.

Step 3- The stone is crushed and mixed with other materials (such as clay). The mixture is heated to 1400 degrees Celsius to produce clinker, which is then ground into cement.

Step 4- The cement is packed and transported to the island by boat.

Step 5- The boat is unloaded and the cement is now loaded to a car/ small truck and transported to the place where the structures are built.

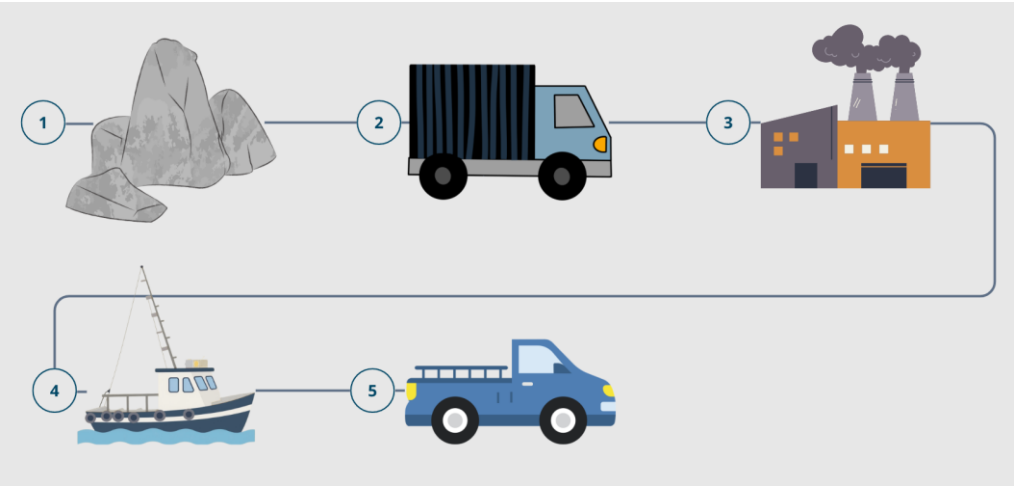


Fig. 11: Activity analysis on how cement is made and transported, graphic elements from Canva (2026).

4.3.5 Material

The analysis in fig. 11 made me realise how many transports and how much carbon dioxide is emitted into the atmosphere when cement is produced. According to Jativa, Ruales, and Etxeberria, (2021) this is mostly due to the production process, when turning limestone into clinker. When interviewing people about volcanic ash I learned that this material would be a more sustainable option, if mixed with the cement, since this would lower the amount of the cement used in the mix. If it would be possible to further minimise the use of Portland cement and reduce transport and heating by using volcanic ash and local materials, carbon dioxide emissions could potentially be reduced even more.

To know for sure if this idea would be possible in practice, it was necessary to look at what was needed to create a substitute for Portland cement on-site and if it would be a realistic solution. My earlier research suggested that a cement-like material could potentially be produced using materials such as seashells, volcanic ash, sand and salt water. (See fig. 12 for materials).



Fig. 12: Materials, from the left: seashells, volcanic ash, sand and salt water. Photo of seashells by George Girnas (Unsplash).

Seashells:

The first material is the seashells, which I learned from my interviews must be collected and heated to a temperature of 900 degrees Celsius. This is a lower temperature compared to Portland cement heating, which according to Jativa, Ruales, and Etxeberria (2021) is heated to 1350-1450 degrees Celsius.

Since a sustainable material is a central focus in this thesis, how the seashells are collected is for me an important aspect to consider. One suggested approach for the restoration workers would be to collaborate with restaurants and locals to collect seashells that are waste from food. That will help create some project awareness and with this contact it would be possible to give information about sustainable fishing and seashell species that should be left alone. In some countries they have a large amount of seashell waste that might be an option for this solution as well (see fig.13).



Fig. 13: Beach full of seashells at Bonaire. Photo by Max Enerstam.

Volcanic ash:

The volcanic ash would have to be milled from stone to ash (see fig. 14 for volcanic stone before the milling process). Which according to my interview with people working with this material, would be possible to do at a factory close to the volcano where the stones are collected. My impression of this information is that it doesn't matter which country the volcano is located at, this process would be possible anywhere, with the right knowledge and tools, but it needs to be done at a factory. This means that the stones must be transported to the factory and the ash transported from the factory, like the cement, but it doesn't go through the heating process.



Fig. 14: Volcanic stone before milling.

Sand and salt water:

Sand and salt water is the third and fourth materials. It would be possible and easy to collect sand from the beach and water from the ocean, the sand shouldn't need any cleaning since salt water will be used in the mix. So, if the structures are manufactured at the beach these two materials don't need to be transported at all.

The main reason for exploring a different material was to research the possibility of using an environmentally friendly and local material, to make sure it doesn't need to be transported to the island. Which will lower the risk of material shortage. It probably won't be possible to find volcanic stones on the beach and mill them there, but since this material is a little bit lighter it might be easier to transport. According to my interview volcanic ash is more sustainable and, according to Förster et al. (2024) study, it might create a better material for the corals. Therefore I believe it is an interesting idea to explore the option of using this material.

When it comes to the seashells it would be ideal if they could be heated at the island, close to the beach. That would reduce carbon dioxide emissions from transport, and it would be more convenient since the material is already on the island. According to my interviews the heating process would require a kiln, which is an oven that can handle temperatures over 1000 degrees Celsius. Would it be reasonable to get a hold of that at a remote island?

4.3.5.1 How to build a kiln

My supervisor has a lot of knowledge when it comes to material and concrete, which was helpful. I have also been researching online and watched Primitive Technology*: Downdraft Kiln* (2023) on YouTube to learn how to make a kiln and Primitive Technology*: Lime* (2018) to learn more on how to burn the shells correctly.

My supervisor and my research helped me realise that a kiln can be built almost anywhere, if you have clay that can handle the heat or if you can dig a hole in the ground. This thesis work started in February and since it was done in Sweden, a country with winter months, it wasn't possible to dig a hole to build an underground kiln. The only option was to build a kiln above ground. Clay can be found in the ground in Sweden but since the ground was frozen it had to be bought. To not have to buy too much clay, bricks were used with clay in between the bricks. This was instead of building the whole thing in clay, as seen in Primitive Technology*: Downdraft Kiln* (2023) video.

I started to plan the work on building my own kiln by writing a list of things I needed (see appendix E) and to make a sketch of the kiln (see fig. 15). The measurements in the sketch was taken from the Primitive Technology*: Downdraft Kiln* (2023) video.

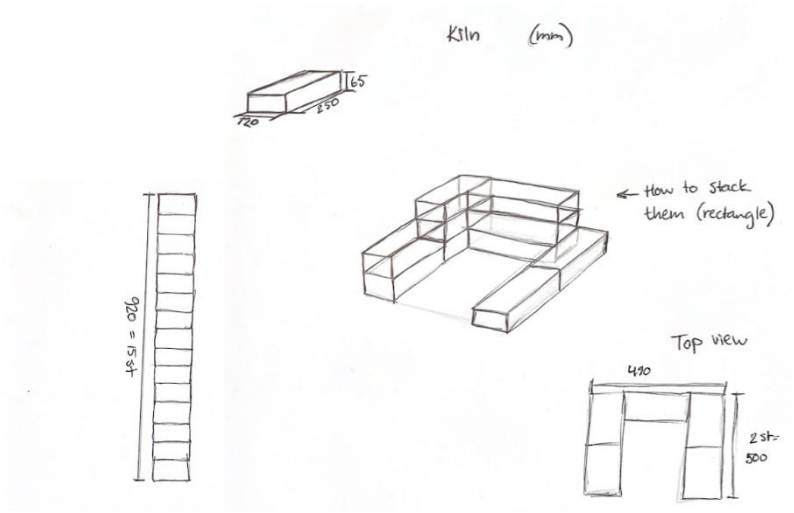


Fig. 15: Sketches on how to build a kiln with bricks.

Summarise define phase:

- Important functions were established after analysing information collected through observations, interviews and a survey.

- Insights like local materials and local manufacturing could benefit the society. It could also help with project awareness.
- A lot is connected to the material within the system. This is therefore an important part of the design.
- It is a complex process with many activities that require human handling.
- A modular structure might be the best option for transports and placing, but it needs to be stable.
- Seashells is a local material that could work for this purpose, but it needs to be processed on spot. Otherwise it might not be practical.
- Volcanic ash sounds like a good option since it is more sustainable and more resistant to sulphates.

4.4 Develop

4.4.1 Exploring a new material

In the YouTube video Primitive Technology*: Lime* (2018) I learned that the seashells needed to go through a burning process before they could replace cement as a binder. The first step of making the material was therefore to try to burn the seashells. To be able to do so it was necessary to build a kiln (see fig. 16).

The bricks were laid out as the sketch in fig. 15 shows and placed on top of each other with clay in between. The clay was carefully pushed out and placed in all the gaps. It is important to build a solid kiln, so the heat stays in the oven, according to the YouTube video Primitive Technology*: Downdraft Kiln* (2023).



Fig. 16: Building an above ground kiln.

After finishing the building of the kiln the clay needs to set and dry. In this case the kiln dried under a tarpaulin for 36 hours before the heating process started (see fig. 17).



Fig. 17: Finished kiln, ready to dry for 36 hours.

The kiln needed to be preheated before burning the seashells (see fig. 18). For this, wood was used. This process took around 3 hours and during this time the kiln was analysed to make sure it did not have any cracks. Where smoke seeped out between the bricks, more clay was added.



Fig. 18: The process of preheating the kiln.

After the preheat, approximately 6,5 kilos of seashells were added on to the kiln from the top, the chimney. The glow was kept to an orange/ yellow colour during the burning of the shells, as recommended by a construction professional interviewed for this study. No thermostat was used (see fig. 19).



Fig. 19: An orange to yellow colour on the glow was kept during the process of burning the shells.

The seashells burned for 24 hours before they were removed with a shovel, small pieces of briquettes were removed from the shells. Most of the seashells had now a grey colour, which means that they have not been burned in the right temperature for long enough. If they burn correctly they get a white colour, according to Primitive Technology*: Lime* (2018). To remove the shells with the grey colour a strainer was used. The shells were shaken in the strainer so the shells that had burned to the finished material, and because of that was fragile, pulverised into a bucket. After this water was added to slake the shells and get the finished material (see fig. 20 for the steps). This was then mixed with water, which created a paste, and stored in a bucket with a lid on. I was told, during my interview with one of the builders, that this paste is called lime putty.



Fig. 20: The process of unfinished shells from finished ones to make lime putty.

In this process I managed to create 1 kilo of seashell powder. I noticed that the weight was somewhat reduced in the process, but with this result most of the shells were not correctly burned. This was most likely caused by low temperatures, so to see if I could get a higher

number of seashells turned into powder, I repeated the process the same day as I collected the first batch of burned shells.

This second time the kiln didn't needed any preheating since it was already warm from the earlier burning of the shells. To get the temperature higher, more air was added to the glow with the help of an air blower. This time I decided to add less shells, approximately 4,5 kilos, so the heat would spread more evenly around the shells. After 36 hours I repeated the process of removing the material from the kiln and strain it. This time I got approximately 1,5 kilos of finished powder, which means that the second time was more successful. I believe that with practice this could probably get even better.

4.4.2 Making and analysing material samples

To test the material and to develop a working mixture, small samples were made with different compositions (see fig. 21). The mixtures were put in aluminium forms, in every form a sample of 100 grams were added. Different material mixtures, with a variation in the amounts of lime putty, volcanic ash, Portland cement, sand and water (both fresh and salt water) were made. (For all the compositions see appendix F).



Fig. 21: Making material samples

The salt water used was freshwater mixed with sea salt, 35 grams salt on 1 litre of water. This was intended to imitate salt water. Based on the interviews the water content was identified as one of the most critical aspects when mixing the materials, especially when the volcanic ash was included. The lime putty was therefore weighed both wet and dry to have a reference on how much water it contained, the weight difference was 55%.

A total of 19 samples were made. As a reference, three samples with Portland cement and volcanic ash were made. This helped understand how the samples mixed with lime putty differed to Portland cement, when analysing and comparing the samples.

The samples were then stored under plastic foil. They were analysed, weighed and sprayed with water once a day during the first week (see fig. 22). The temperature in the room was

checked every day, it varied between 19.8-20.8 degrees Celsius. From day seven and forward no more water was added and the tests were now analysed once a week instead. This analysis process was suggested by professionals in the construction industry during interviews.

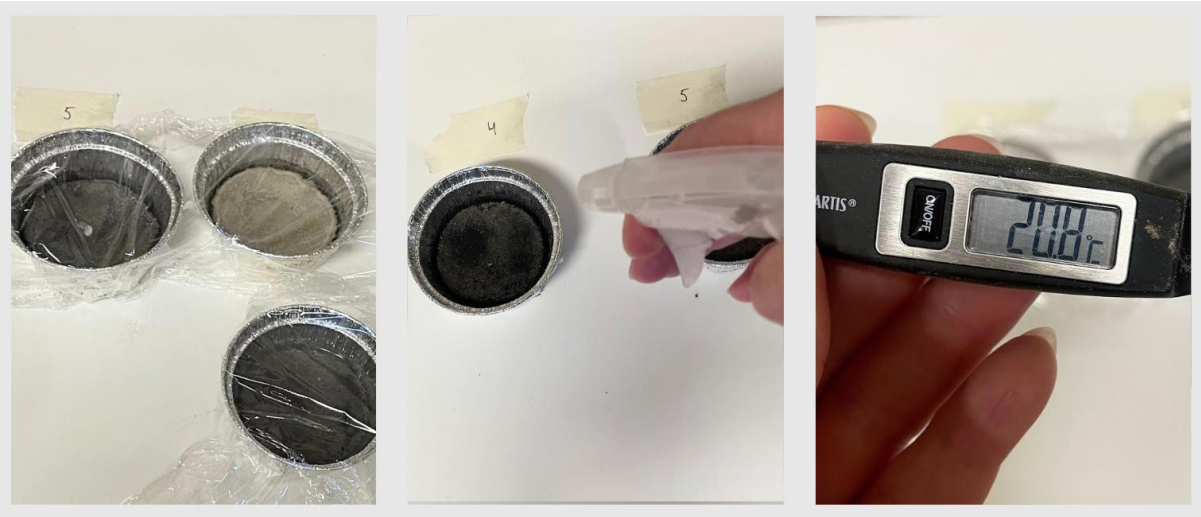


Fig. 22: Samples stored and analysed.

After 3 weeks of letting the samples cure, it was time to remove them from their forms to analyse the finished sample. Many of the tests did not make it after they were removed from the form. This result indicates that the proportions of binders and sand are critical to create a strong material. Samples with high sand content failed, due to the lack of something that can bind all the sand together, this is shown in fig. 23, where two samples are compared. They contain the same amount of cement and volcanic ash, but sample two contains much more sand and therefore can't handle the pressure of one finger. (For mixture compositions of the analysed samples see appendix G).



Fig. 23: Comparing samples. Sample to the left can handle salt water. Sample to the right, with more sand, can't handle the pressure from a finger.

Samples with lime putty were generally weaker than the ones containing Portland cement (see some of these samples in fig. 24). Could this mean that lime putty requires a more precise balance of the ingredients to work effectively as a binder?



Fig. 24: Samples containing lime putty were generally weaker than samples containing Portland cement.

The most successful mixture, that included lime putty, was a hybrid between cement, lime putty, sand and volcanic ash (see fig. 25). This indicates that a mixture containing lime putty require one additional binder. Of all the samples made, some were mixed with salt water. An observation among these samples was that the use of salt water appeared to increase the material strength. Even the samples that contained salt water and failed needed more pressure to crack, than the sample with the same mixture but that was mixed with fresh water.



Fig. 25: These two samples are both a mix of all the binders. The one to the right contains salt water, which seems to make it even stronger.

In total 7 out of 19 samples remained after removing the form. These were put in salt water and analysed once a week for three weeks. During this analysis every sample was removed from the water and was exposed to pressure from hands. The surface was scratched and analysed as well. After the first week the samples with fresh water and a higher amount of lime putty started to dissolve on the surface, they still had a strong core, but they were not going to make it (see fig. 26). After this, 4 possible samples remained, which of 3 contained a minimum of 20% volcanic ash. These mixtures will be explained further in the “execution-deliver” part of the thesis.



Fig. 26: Three samples started to dissolve after one week in saltwater.

4.4.3 Brainstorming

To create ideas for the shape of the design multiple brainstorming sessions were done in groups with other students. The sessions differed depending on how far into the design process the project had progressed. The focus remained the same, to get insights from others point of view and to create a lot of ideas. Which according to Wikberg Nilsson, Ericson and Törlind (2021, s125-137) is the goal with a brainstorming session.

During the first session the group discussed coral reefs and how to create a pattern with the structures, since multiple structures will be placed close to one another. Sketches of possible shapes that would be convenient to transport in the boat and under water was made (see fig. 27). These sketches were then analysed and discussed with my supervisor and one of the construction workers I earlier interviewed. I then realised that none of these first ideas would be convenient to manufacture, since they have some small and fragile parts. They would also be hard to stabilise on the seabed since they all have a flat base. How the ideas were analysed will be presented later in the PNI part of the thesis.

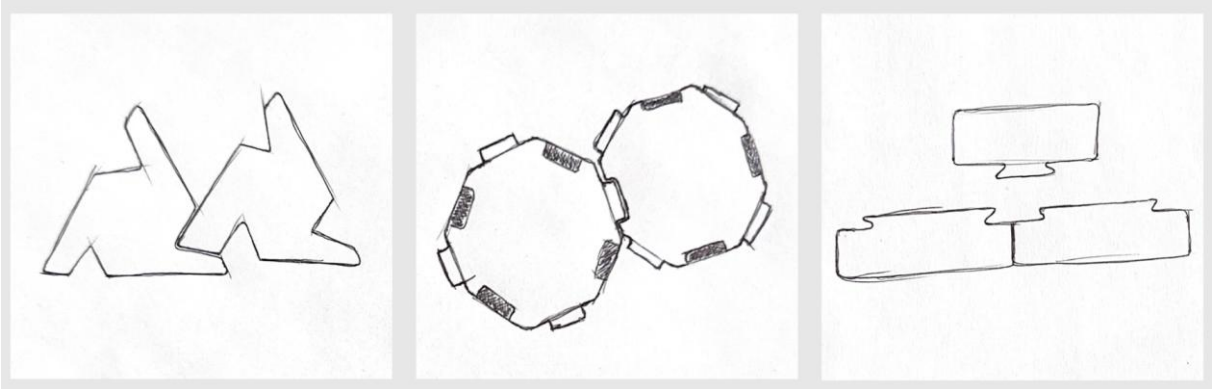


Fig. 27: Sketches after brainstorming in group.

After analysing and reflecting over my first ideas, I came to the realisation that a modular design might be the most convenient solution. Such a design could address the user's needs

with manufacturing, transportation and placing of the structure. This was based on the assumption that a structure that can be disassembled is easier to pack and transport. Discussions with my supervisor also suggested that it would be easier to manufacture the parts if they had a simple geometry. In addition, I envisioned how the different parts could have functions that met the different needs shown in the function analysis in Fig. 6.

During this time in the project, a different brainstorming session was done with a new group. At this stage the possibility of a modular structure was investigated. I needed to explore how to make the structure stable, once placed, as well as how different parts could address different needs. Therefore ideas on different modular concepts were generated in this session (see fig. 28). For 3 minutes all the ideas that came up were written or sketched down on one paper, with a theme written in the middle of the paper. The themes used during the session were: Things that are put together, things that support each other and how to staple things. As said, the themes were chosen to generate ideas for a modular structure, where the parts could stabilise and create a higher weight together once placed. To visualise some of the ideas in three dimensions, clay was used to make quick 3D-sketches. This session provided inspiration to explore new ideas, but it did not lead to a practical solution.



Fig. 28: Brainstorming on different themes in a group.

4.4.4 Analogies

At this stage of the project, I had a hard time to create ideas that could met all the user needs and at the same time be stable enough, once the structure was placed. Therefore, the analogy method was used to generate new ideas and to approach the problem from a different perspective. With the help of different analogy themes ideas were sketched down during 3 minutes on each theme. Wikberg Nilsson, Ericson and Törlind (2021, s127) recommends the designer to use both analogies close to the field of the design and far away from it. They also emphasise the importance of divergent thinking to generate a broad selection of ideas. One of the analogies used during my session, the only one that provided new and relevant insights,

was: what if the solution was an iceberg. This analogy is somewhat close to the subject tropical reefs since both reefs and icebergs exists in the oceans, but it is still far away from it. This analogy led to floating transportational ideas (see in fig. 29). After the session the ideas were analysed with convergent thinking to see which of the ideas that were realistic. The floating ideas would probably not be reasonable or safe solutions, if performed exactly as the sketch in fig. 29 shows. But attaching the structures to a floating device might be a good idea for under water transportation. If this can help the divers transport the structure from the dropped spot to where they should be placed, could that save time and would it make the work easier?

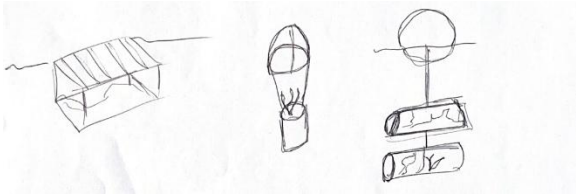


Fig. 29: Fast sketches of ideas using the method analogies.

4.4.5 Prototyping

To be able to test and analyse ideas, 3D models in different materials were created. In the earlier stage of the project, ideas from the brainstorming sessions were analysed by making clay models. Clay was a good option to work with since it is possible to change and test ideas fast, see fig. 30.

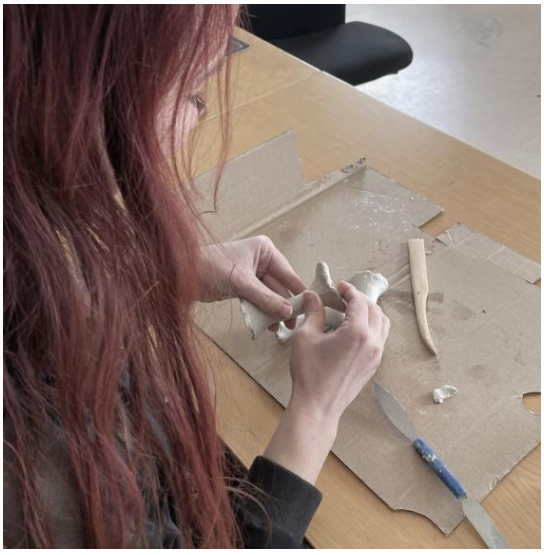


Fig. 30: Making clay models.

After the brainstorming and analogies sessions I initially found it difficult to generate relevant ideas. Midway through the project, a presentation of the work was held. The feedback I received after the presentation provided me with new insights. This led to the realisation that the structure would be much more stable if it had three points of contact with the sand. To further explore this idea, a prototype session was conducted with a friend who is both a construction worker and a diver. Small- scale prototypes were made using sticks and

clay. The prototypes were then analysed and improved. According to Norman (2013, s228-229) testing prototypes with the user is essential to get new insights during the design process. The session with my friend provided this. Particularly since safety concerns were addressed, such as avoiding parts that are sticking out from the structure that the diver could become entangled in. After this session I made some sketches to develop the ideas even further (see fig. 31).

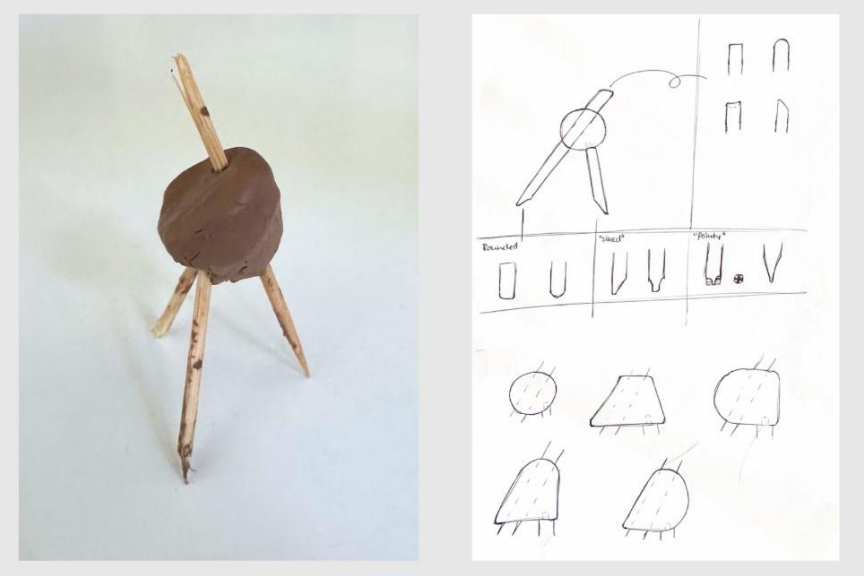


Fig. 31: To the left is a prototype made of sticks and clay, to the right is sketches on how to develop the prototype.

After reflecting and analysing these ideas, 5 stable and modular concepts with focus on the possible to manufacture and transport were made (see fig. 32).

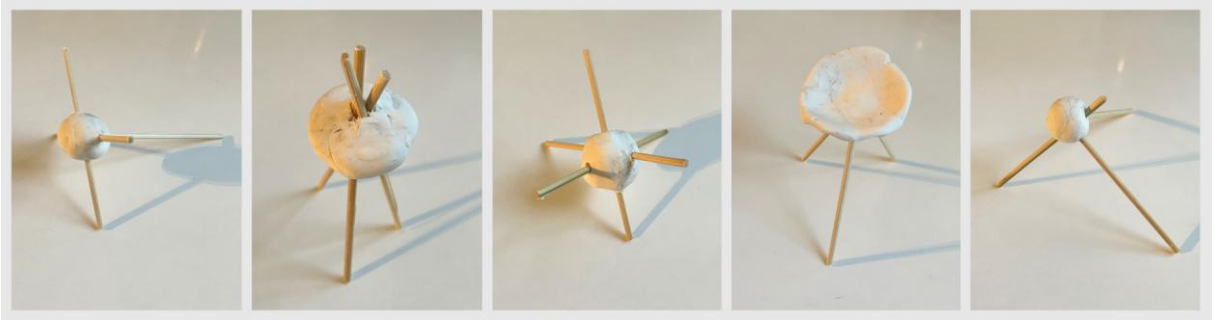


Fig. 32: 5 stable and modular concepts that focus on the possibility of manufacturing and transport.

To validate these 5 concepts, I asked a construction worker, a diver and a gardener to look at my prototypes and say something positive, negative and interesting about every prototype. This method is called PNI and will be explained in the next part of the thesis. This validation, together with a selection process in which the structure's strength and weight were considered, led to the final shape.

Once the shape was established it was time to analyse the design at full scale, to see the proportions and to analyse different measurements (see fig. 33). This full-scale prototype was made in carboard.



Fig. 33: Prototype in cardboard, comparing measurements of the legs.

4.4.6 PNI and selection

The method PNI was used to compare and validate different ideas during the process. Positive, negative and interesting parts on the ideas generated in the brainstorming and prototype sessions were written down (see fig. 34). This helped the process of analysing and validating the ideas (see appendix H for PNI on more ideas).

The PNI method was conducted at two different stages of the process. From the first brainstorming session, how different shapes could create patterns were explored. I realised that shape needs to be simple and stabilisation is hard if the structure has a flat base. Because of PNI and analysing the ideas I realised that a modular structure would be easier to transport and the parts could have different shapes to meet different needs.

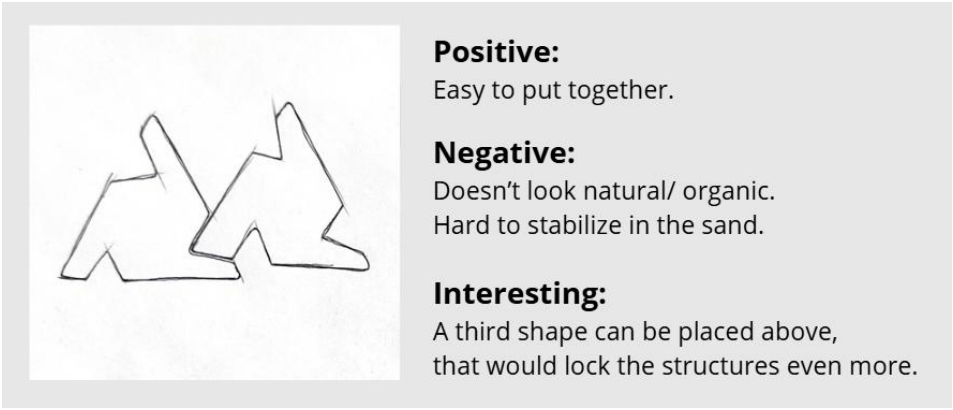


Fig. 34: PNI on different ideas.

After multiple sessions of generating and validating ideas I generated 5 concepts. After building small prototypes on these concepts they were validated through PNI by a construction worker, a diver and a gardener (see fig. 35).

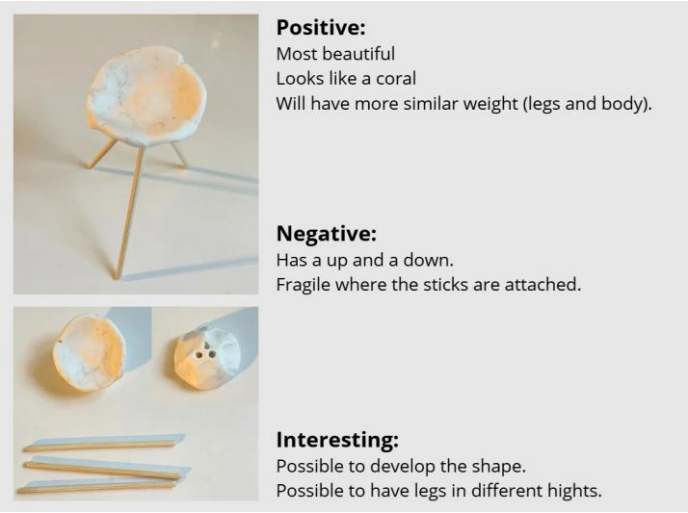


Fig 35: PNI on one of the 5 concepts.

After this, 4 concepts remained. To narrow it down to one idea a selection was made, here the shapes strength was discussed and estimated with one of the construction workers I earlier interviewed. The weight on each structure was approximately calculated by analysing the volume of a full-scale model in an CAD programme (see fig. 36). The difference between the PNI and the selection was that other aspects were added and considered in the selection. Here all the ideas that created unrealistic expectations on the restoration worker, such as too heavy solutions, were removed until only one design remained.

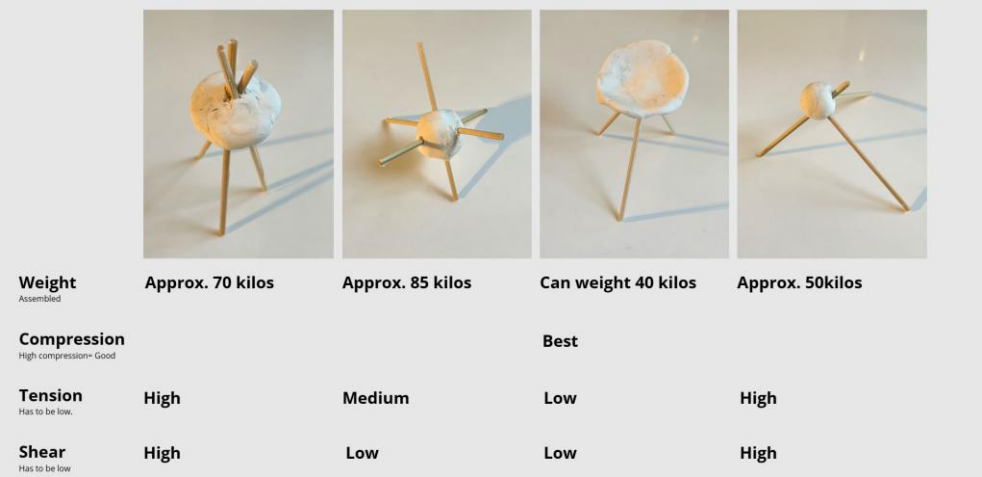


Fig. 36: Selecting one idea out of 5 concepts.

Summarise develop phase:

- The process of burning seashells involves a learning process, with results improving through experience. For consistent quality someone needs to be trained. This could generate jobs locally.

- More material samples need to be made and tested. Samples mixed with salt water indicates a stronger material.
- Analysis of material samples indicates that a percentage of Portland cement in required in the mixture to achieve higher strength.
- Volcanic ash in the mixture seams promising. To ensure a material more resistant to sulphates a minimum of 20% volcanic ash needs to be in the mixture (information gathered in interview), my result indicates that this is possible.
- The structure needs to incorporate multiple functions in order to meet the users needs. A modular design could provide a solution to this.
- A floating device could help the underwater transport of the structures.
- A stable structure needs three points of contacts to the seabed.
- Prototyping with someone who knows the field/ a user can give new insights.
- It is valuable to validate the design with users.
- Prototyping in full-scale can generate insights of proportions and measurements.

4.5 Deliver

4.5.1 Material samples

After three weeks in salt water, a final analysis of the material samples was conducted. Four samples remained intact (see fig. 37). Sample 1 and 2 in fig. 37 contained only cement and volcanic ash. The difference between them was the amount of volcanic ash added, sample 1 contained 10% ash and sample 2 30%. No visible difference could be identified between these two samples.

Sample 3 and 4 contained a hybrid mixture of lime putty, volcanic ash, cement and sand. The main difference between these two was the amount of lime putty and cement they contained. Sample 3 included more cement than sample 4, sample 3 was also the one that contained the highest amount of cement out of all the hybrid mixtures (see appendix F for full list of all the compositions).

According to this study, Portland cement appears to be overall a more suitable than lime putty for moulding the structures. However, the hybrid sample number 4, with a higher amount of lime putty also remained stable during the test period. The uniqueness with this sample was that it contained salt water instead of fresh water.

This could indicate that lime putty has the potential to partially replace Portland cement, if mixed with salt water and volcanic ash.



Fig. 37: Finished material samples.

4.5.2 Full size model function model

At this stage of the project, it was necessary to develop a function model in a stronger material than cardboard. The reason for this was to be able to interact, analyse and visualise a model more similar to the solution, which is an important part in research through design according to Stappers and Giaccardi (2026). This could help demonstrate how the structure will be transported and assembled. To visualise the handling of the structure throughout all stages, details supporting the structure's function were included in this model.

To deliver a full-scale model on time that could be assembled and disassembled, the model needed to be built in a different material than moulded in the concrete-like material I had composed. This was because the material would not have time to cure before the project deadline. My supervisor therefore suggesting constructing the model from wood, fir-tree, and XPS foam (an orange-coloured modelling foam). These materials were not too expensive and could be shaped to resemble the intended form of the moulded structure.

Even though it is time to deliver a finished model and the shape is set, it is still important for a designer to work analysing and reflective (see fig. 38 for sketch of the intended design before the building of the model started).

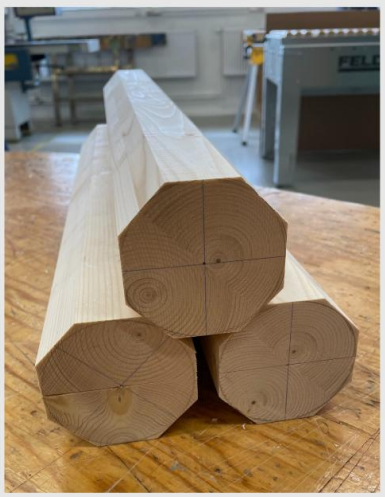


Fig. 38: Sketched of intended design, before the building started.

While making the legs for the design, that was supposed to be cylindric, it was necessary to first make them into an octagonal form due to how the wood material works. When the legs were in octagonal shape, I started to staple them onto one another and test how they would stay in that position (see fig. 39). I realised that this shape remained in place and therefore would be easier to transport by boat. In contrast, a cylindrical shape would be more likely roll around during transport. The octagonal shape therefore became a part of the final solution.

The legs end in a diagonal cut, which is intended to help with the stability. By rotating and adding some pressure from above, a diver can drill down the legs into the sandy seabed. This function was developed by analysing the prototype in cardboard. Testing this with the wooden prototype provided insights into how the octagonal shape also improved grip during the rotational movement.

The legs are attached to a hole in the top part of the structure by a cone-shaped top. When the cone was lathed, I realised that it had to be higher and attached further into the top part. This was needed to create more stability and for the diver to be able to see more of the cone from above, to make the assembling easier, see fig 39 for building process.



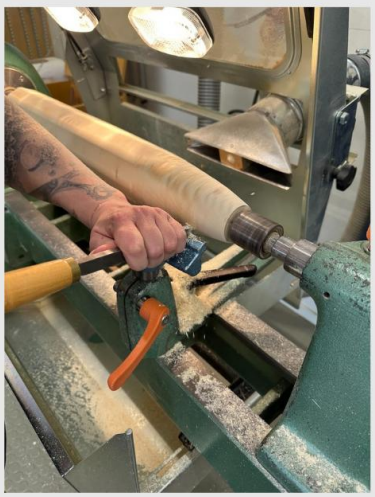


Fig. 39: building the legs for the full-scale model.

The top part of the structure was built in the XPS foam (see fig. 40 for the building process). After creating four circles that were placed on top of each other, I realised that the top part would be too high with the measurements I produced (hight 20cm X top diameter 35 cm). The parts need to be easy to carry, and it is crucial that the structure is stable once placed. With this measurement the structure will be hard to carry, because of the angle that the hight is creating and the currents will have a lot of material to hit on the structure. It was therefore necessary to change the hight of the structure. This also led to a change in the diameter, otherwise it is a risk that the structure would be to light. So, with the new measurements, hight 15cm and diameter 40cm, the structure ended up flatter then intended from the start, which will help in both the carrying and the stability.

The top has three inlets, which will help with the grip while carrying the structure. When a current hits the structure the angle of the top part is intended to push the water down, so it disappears under the structure. An organic form is explored on the top, this was mostly intended to be an aesthetic form that will make the structure resemble a coral more closely, but it might also help the water flow on top of the structure.



Fig. 40: building process of the top to the full-scale structure model.

The structures shape is now finished, and a full-scale model is presented in fig. 41.



Fig. 41: A full-scale model with a finished shape.

4.5.3 3D-printed models

To explain how the structures will support each other, by standing next to one another, and to visualise how a smaller reef could look like, 3D-printed models were created. By using a CAD program, a designer can generate a model that can be printed. In this thesis seven models were printed in white coloured plastic to create a white model of a small reef.

First one model was printed and analysed (measurements for the top part was: height 30cm X a diameter for 50cm). It was clear that the top part was too big for the legs with that first model, this was before the building of the full-scale model began (see fig. 42). The measurements were therefore changed before new models were printed (see fig 43).



Fig. 42: 3D-printed model with wrong measurements.

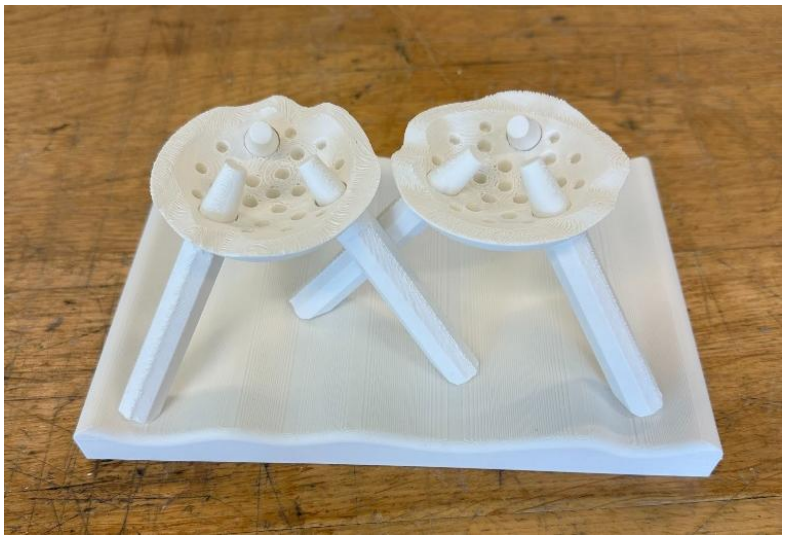


Fig. 43: 3D- printed models with correct measurements.

4.5.4 Scenario

Since the solution was developed to support all the stages of the restoration process, from material to planting of the corals, it was essential to visualise all the steps. Therefore a scenario was made, which is a proposed sequence showing how the user interacts with the solution throughout the process (see fig. 44). In a CAD program a moulding form was created together with other assisting tools needed for the process, such as a rope and tension bands. The model and the assisting tools were then rendered and implemented in an environmental picture of the intended situation. This helps visualise the finished concept and the steps of the process for the user.

The proposed process: First the structure is moulded, this happens at the beach, close to the spot where the structures are then dropped. The moulded parts are loaded on the boat and then dropped at the spot. The divers can then collect the parts and start to assemble them. The picture to the right in fig. 44 shows an assembled structure that corals can be planted on.



Fig. 44: Scenario of the finished solution. Renderings have been photoshoped in a background photo to visualise how the product is used. Background photo 1 by Rio Andhika (Unsplash), 2 by Erwan Hesry (Unsplash), 3 and 4 by David Boca (Unsplash).

Summarise deliver phase:

- No visible negative aspects of a concrete made by Portland cement and 30% volcanic ash is shown.
- A hybrid material mixed with salt water has a better resistance to salt water then the same composition mixed with fresh water.
- Prototyping in full-scale can give new insights to the shape and function of the design.
- Measurements need to be analysed and reflected on a full-scale model.
- It is important to analyse every step of the design process, test the model as much as possible.
- An octagonal shape provides both stability and grip.
- A scenario can help visualise the result.

5. Result

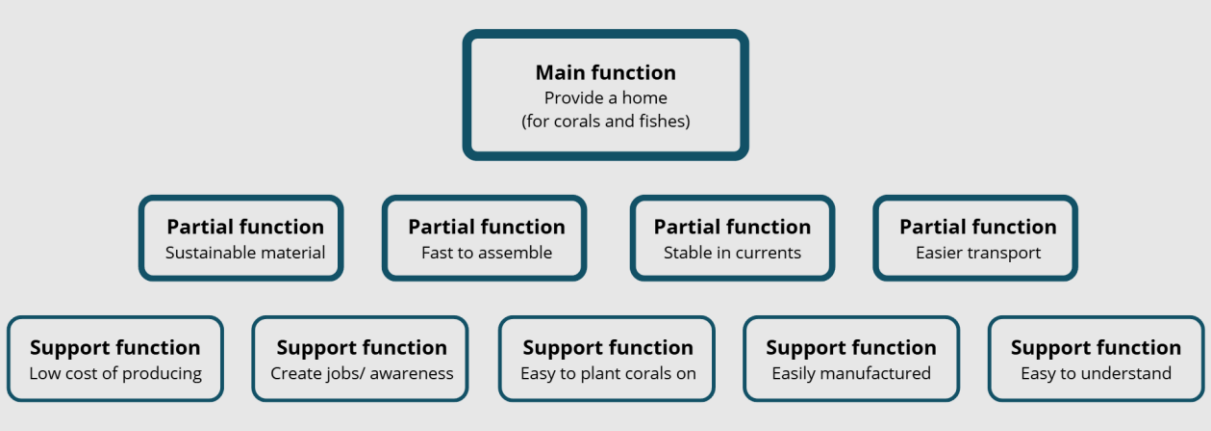


Fig. 45: Function analysis, repeated for reference in the result chapter.

5.1 Sustainable material

In this thesis material has been a central part of the work and the focus was to investigate a more sustainable material, that can be found on or close to the spot where the structures are going to be placed. Since concrete is a common material to manufacture the structures in today, but it is not the most sustainable material and, in some cases, hard to find locally, a concrete-like mixture was developed and tested. The goal was to examine whether waste or local materials could be used to mould the structures in, and if the material could be compared to the Portland concrete, when it comes to resistance to pressure and from salt water.

Seashells were therefore burned to create lime putty, which were tested as a partial substitute for the clinker, the main binder material in cement. This was mixed with volcanic ash, sand and water, materials that can be locally found. Multiples samples were made and analysed. The results involving self-made lime putty in this study showed that the material does not have the binding effect that is necessary to create a long-term stability for the structure. Therefore, it won't be reasonable to eliminate all the cement, since this material provide a structural strength to the finished result that can't be replaced with the material substitutes tried in this thesis.

However, the mixtures where volcanic ash replaced an amount of Portland cement (10-30%) remained stable throughout the entire test period, and so did two of the hybrid mixtures, which included all the mentioned materials (see fig. 46 for finished samples). Since volcanic ash is a much more sustainable material than Portland cement, this result indicates that a more sustainable material can partially replace Portland cement while maintaining structural stability. When it comes to using waste/ local materials the result of the hybrid mixtures indicates that if the material is mixed with salt water, not fresh water, it showed better resistance and durability in salt water. The hybrid samples showed that the amount of Portland cement could be reduced by approximately 50% (roughly counted on how high percentage of Portland cement that was used in the two samples).

The material experiments focused on the partial function- sustainable material in the function analysis (see fig. 45 for function analysis).



Fig. 46: Material result. First one from the left: cement mixed with 10% volcanic ash. Second one: cement mixed with 30% volcanic ash. Third one: 50% cement, 20% volcanic ash and 30% lime putty. Fourth one: 35% cement, 20% volcanic ash and 45% lime putty (mixed with salt water).

5.2 Moulding and manufacturing

The result is a modular moulded structure designed to be manufactured at a spot close to the beach (see fig. 47). The manufacturing process requires the material mixture, moulding forms and tools for mixing and holding the mixture before it is poured into the forms. The four parts of the structure are moulded separately in forms that can be 3D-printed. Before casting, the inside surfaces are coated with oil or grease to make the cured parts easier to release.



Fig. 47: Proposed manufacturing process at the beach. Background photo by Rio Andhika (Unsplash).

The moulding form for the legs has an opening at the top where the material is poured in. As shown in fig. 48 the form is angled in a way that creates the diagonal cut at the end of the leg during the curing process. This means that the shape is established directly in the mould

without additional processing. To remove the cured leg lids are opened from both ends of the form, and the finished part can be pushed out using a tool such as a small wooden plank (see fig 49).

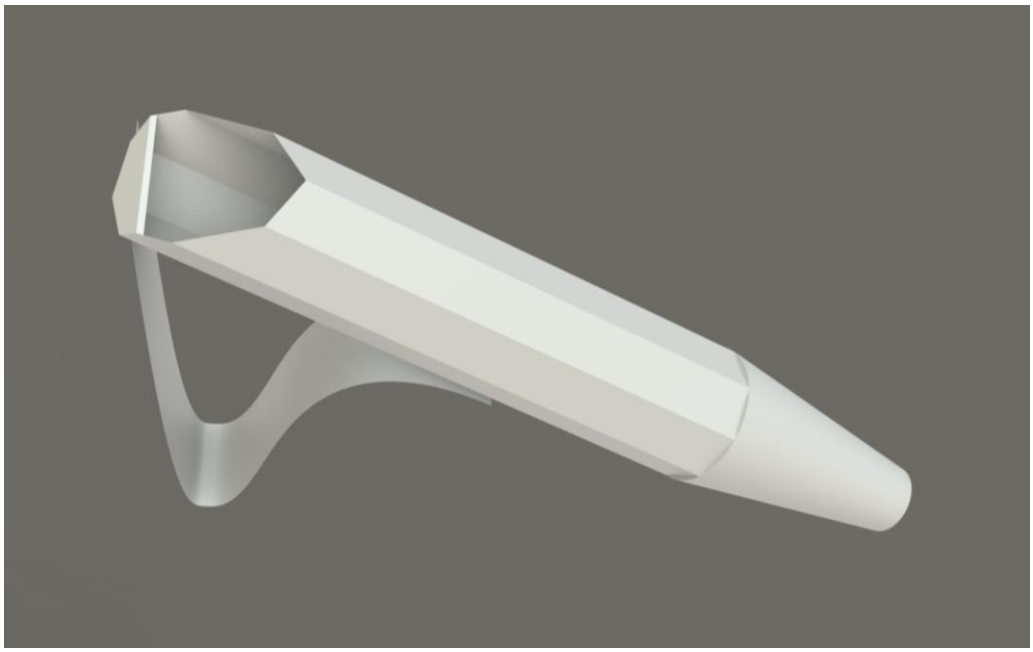


Fig. 48: Moulding form to mould the legs of the structure.

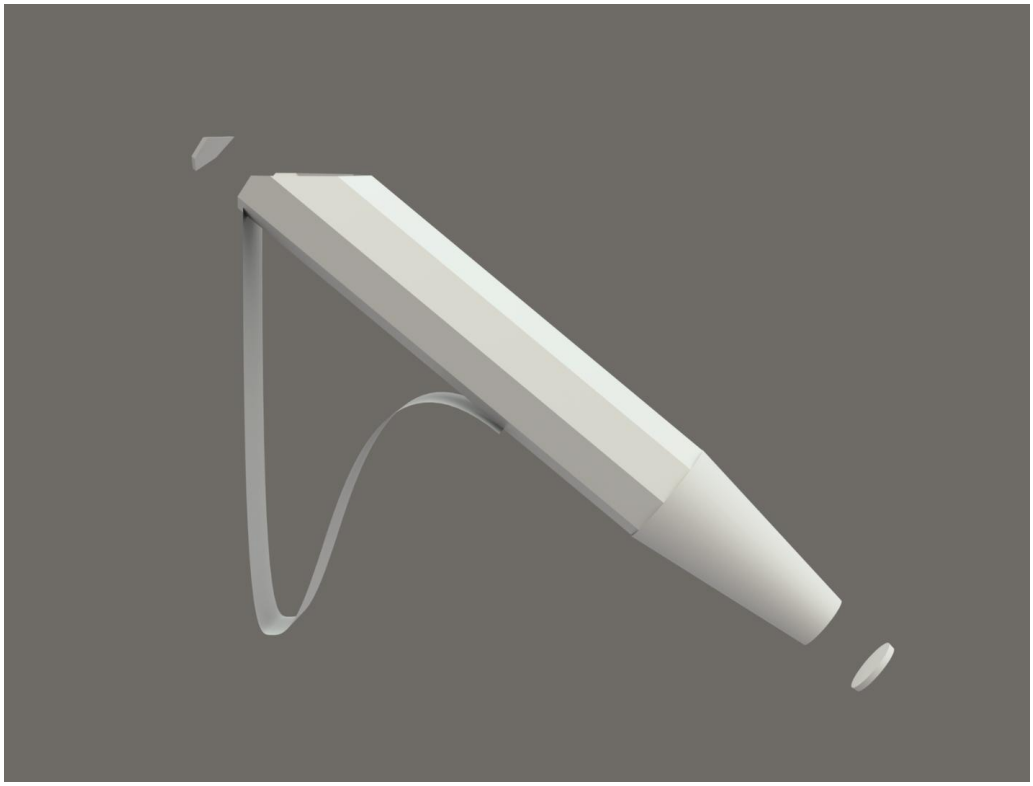


Fig. 49: The lids are opened to remove the finished part.

The top of the structure has two forms, one creating the outer shape and one shaping the inside (see fig. 50). After brushing the forms with oil or grease the material is poured down in the outer shape until the line indicating that enough material is added. The inner part is then

pushed down into the material, making the material level rise to the edge of the outer mould, (see fig. 51).

The inner part creates the holes used for coral planting and well as the soft wave-like shape along the edge of the structure. It is important not to let the material settle too much before the inner part is added and to ensure the two moulds align correctly to create the holes for the legs. To help position the moulds correctly, the inner part has three guiding holes that fit into the leg's connectors of the outer mould.



Fig. 50: Moulding forms to mould the top part of the structure in.



Fig. 51: Visualises how the inner part is pushed down in the outer.

This manufacturing method makes it possible to produce multiple structures using the same mould, which becomes cost-effective when producing larger quantities. It also makes it easy to manufacture them on-site. The curing time of a mixture including volcanic ash would be closer to 28 days, this result will therefore require multiple moulding forms. This result

relates to the support functions- easy to manufacture and low costs of producing in the function analysis (see fig. 45).

5.3 Transport

The exploration of the new material was partly to minimise the amount of transport required for the material to get to reach the restoration site. With this design proposal it won't be possible to remove all the transportation, since the volcanic ash needs to be transported to and from a milling factory. The result of this study has also shown that cement is a necessary material, which will also have to be transported to the island, probably a lower amount than before but it is still needed. The remaining materials should be possible to source locally on the island.

Once manufactured, the legs are secured in a package of three using a tension band (see fig. 52). The top part of the structure can be secured using a rope threaded through the three leg openings. How the transport is conducted will probably differ from site to site, sometimes a small boat will transport the structures and drop them in the ocean and sometimes the divers will bring them from the shore. In the first scenario, the structures can be lowered down by using ropes attached to the tension band. If the divers transport the structures directly from the shore, the parts may be attached to a floating device such as a buoy. This could reduce the physical strain on the divers since most of the weight would be carried by the buoy while the structure is pushed or guided through the water. A medium-sized buoy with a diameter of 50cm could carry approximately 50-55 kilos, which is more than the weight of three structures in salt water (approx. 42 kilos).

The same principle could be done when the structures are dropped by boat, allowing the divers to transport them from the drop-off spot to the placement site more easily.

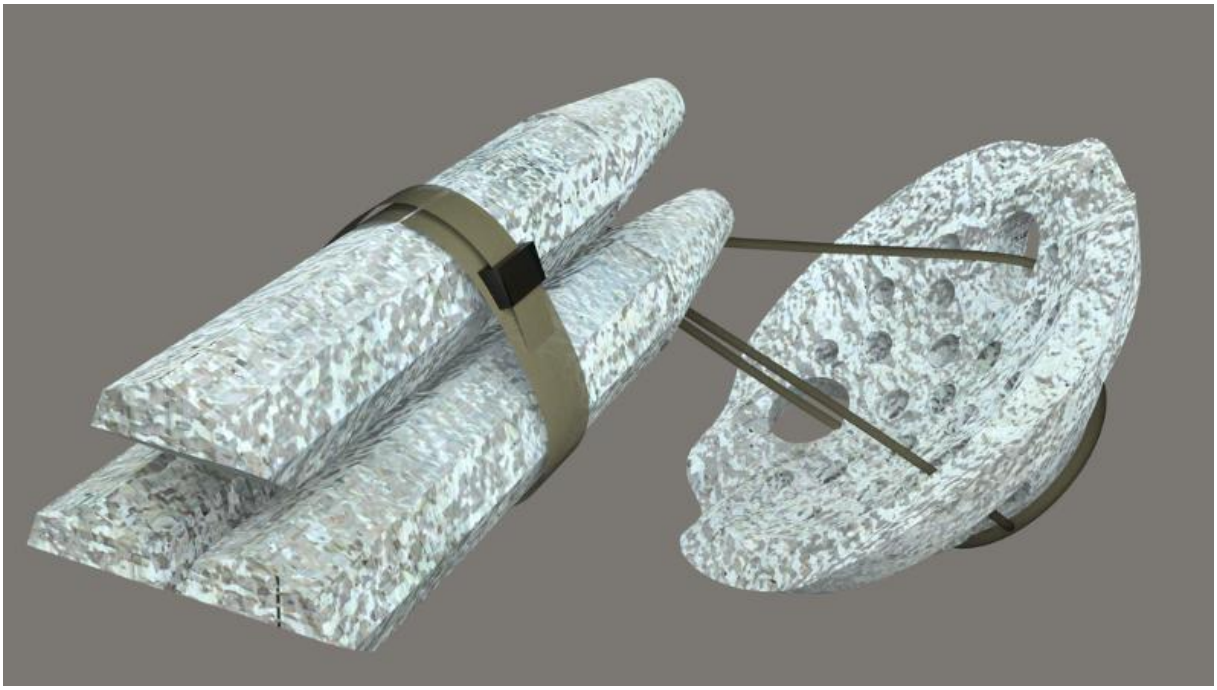


Fig. 52: Transport of the manufactured structures.

The modular design and the shape of the structures will make transportation easier. The octagonal legs and the possibility to transport the top part upside down were design intentions that focused on the partial function- easier transport, of the function analysis (see fig. 45). The structures are also designed in a shape that makes it possible to attach them to some sort of floating device which may further simplify transportation and handling.

5.4 Assembly and placement

Based on personal experiences it could be wise to split the team up into two groups once assembling the structures. One group will handle the transportation of the structures under water, and the other group will stay at the placement site and assemble the structures. The transporting group will remove the tension band and the rope from the structure legs and leave them on the sand for the assembling group. It requires one diver to assemble one structure, and the process is as follows (see fig. 53). First, the diver takes one leg and slightly pushes it down in the sand. The top of the structure is then placed onto the leg, with the cone fitted into one of the holes. If the environment is calm the structure can now balance on its own. The second leg is then attached to the top part. To place the final leg the structure needs to be tilted towards the two legs that are already positioned. Once assembled the legs can be rotated to drill the structure further down in the sand.

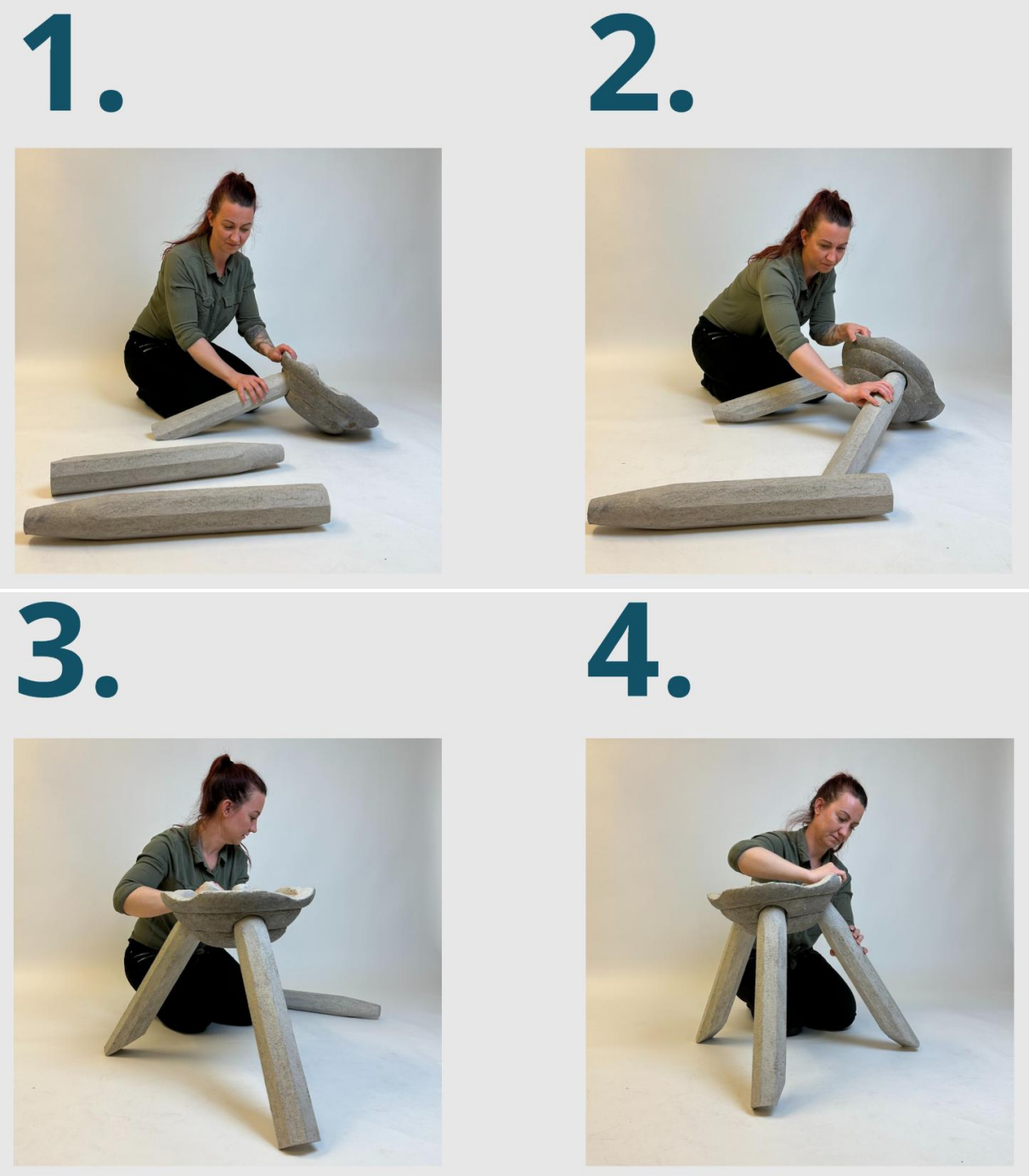


Fig. 53: The assembling process of a structure.

Several aspects of the structure's design are intended to simplify the assembling process, which relates to the partial function- easy to assemble in the function analysis (see fig. 45). The diagonal cut at the end of the leg will help dig down in the sand if the leg is turned. The cone-shaped top of the leg allows it to lock into the structure at multiply angles. The inlets on the top part of the structure will help with the grip while carrying or holding the part. The legs are easy to attach since you can see from above where they go and the hole has a wide countersink that will help guide the leg into position.

Without field testing, it is not possible to determine whenever the structure fulfils the partial function stable in currents, or whenever first-time users will find the assembling

process easy, as described in one of the support functions. However, the function model remains stable on uneven surfaces since the weight from the top part is pushing the legs downward and outward (see fig. 54).



Fig. 54: Function model placed on uneven surface.

5.5 Coral planting and attracting coral larvae

The result has multiple holes in different sizes to make it easier to plant coral fragments by attaching them within the openings (see fig. 55). The varying cone shaped openings are intended to help secure the coral fragments in place. In the research phase of this project, it was known that the coral larvae prefer texturized surfaces to attach. Therefore, the moulding forms are intended to be 3D-printed, since previous studies have indicated that the coral larvae respond positively to a 3D-printed surface. Moulding the structures in a concrete or concrete-like material also contributes to a more textured surface.

In theory, the use of volcanic ash in the material could help attract coral larvae and provide nutrients that may improve coral resilient. This was not tested within this thesis and is instead based on findings from previous studies.



Fig. 55: Holes on top of the structure for coral planting.

The structures are placed close together to support one another (see fig. 56). This creates a larger space underneath the structures, where observation of existing structures has shown that larger fish can find shelter. The parts of the legs extending above the top is intended to function as a hiding place for smaller fish, and before corals are planted in the holes, they can also provide temporary shelter. If weight is put on the top part of the structure the structures legs get pushed down and out. Therefore, if turtles were to use the top surface as a resting place, the added weight could potentially help press the structure further down into the sand. These functions support the main function- provide a home for fishes and corals, of the function analysis (see fig. 45).

The structure is therefore designed to support both coral growth and marine life, with the intention of contributing to increased biodiversity.



Fig. 56: 3D-printed white model that shows the shape that the structures create together.

6 Conclusion and discussion

6.1 Material and sustainability

To have a sustainable design approach it is important to consider the system as a whole, suggests Ceschin and Gaziulusoy (2016). They also emphasise the importance of implementing social and economic sustainability in the process for a more reliable result. This approach is something that has had a focus during this project. The full process, from collecting the materials to a finished and placed structure has taken into an account. The search for a more environmentally sustainable material has been at focus, but aspects like cost and what this material could do for the society has taken into an account and been reflected on.

About the material result: I think that coral reef restoring is in itself a sustainable initiative, and the lack of a completely sustainable and locally sourced material should not prevent restoration work from being carried out. However, it is still important to reflect on the materials that are used today and to explore alternative options. The result in this study indicates that there are more sustainable material approaches that can achieve a strength comparable to using only Portland cement.

The self-made lime putty is a locally source the material from seashell waste. It is a process to create this material, similar to the process as creating clinker for cement, but the approach here is to create it on the restoration site by yourself. The process requires both time and practical knowledge, meaning that it is necessary for the restoration workers to have a good relationship with the locals to help collect and burn the shells. This approach may therefore create project awareness. My second process of burning the shells resulted in a higher success rate then the first process. This indicates that experience and training are

important for archiving consistence material quality, which could therefore generate jobs in the society. This process might not be reasonable for every restoration site, but if it is it could give a positive outcome on all three aspects of sustainability.

The kiln in this study was built in the end of March/ beginning of April in Sweden. The ground was still frozen and the air around 5-10 degrees Celsius. It was therefore required to build the oven above the ground and with bricks to create the heat that is crucial for the burning of the seashells. Looking at the history, it would be possible to get the same result without a kiln above the ground if one would dig a pit and create a kiln underneath the ground, which requires a lot less materials and could be done without special tools, such as bricks and trowels. This could unfortunately not be tested in this study because of the location and weather in that location, but it shouldn't be a problem for the locations where the structures are supposed to be created.

One of the most interesting findings in this material exploration was the samples mixed with salt water performed better than those with fresh water. This could be much more cost effective, since salt water is found locally and free on the islands. It is also more sustainable than using freshly cleaned salt water. This is worth exploring further in the exploration of new materials.

6.2 A modular solution

The starting point of this thesis work was a problem identified during field work and observations of a coral restoration team. The structures were difficult to handle under water because of their shape and weight. For example, their legs were pointed outwards from the centre forcing the divers to carry the structures at a distance from the body, which made the carrying heavy and reduced the control under water. It was clear that the structure was designed to stand as a substitute for the non-existing reef, not to be handled along the way. As an industrial design student, I have learned that things can be designed around humans and their interaction with the design. This does not have to change the purpose of the design, the focus can be only about the handling of the product. It could in fact make the design even better and serve a bigger purpose than the initial design did.

During this thesis a human centred design approach was added to the design process. This according to Norman (2013, s221-235) should include four activities, observation, idea generation, prototyping and testing. These activities should involve the user, the designer needs to understand the user and their needs. Norman (2013) says that the observation and testing should involve the user and the process should be iterative. During the process I conducted all these four activities. However, the opportunity of testing the product was limited. Observation, prototyping and validation was done together with the user. The result of this study is a modular design based on a human centred approach. The purpose was to make the handling easier for the restoration workers while keeping the main function of the structure, to give a home for fish and corals. The modular result shows a design where the legs and the top part are carried out separately and assembled under water. This makes it possible for one diver to carry the parts and assemble the structure in an easy way. The shape

also improves transportation and potentially increased stability on the seafloor. This indicates that a design centred around human interactions with it can also give a positive result for the main function.

I believe that a last observation of the user, handling the design on spot could have developed the design details even further, but unfortunately that could not be done during this project. This would probably have led to new insights in the process and made the finished design even more aligned with the human centred design approach.

Without realising it from the start, this thesis adopted a research through design approach. According to Stappers and Giaccardi (2026), research through design generates knowledge through the process of designing, visualising and reflecting on the product. In this case, the design process contributed insights and identified challenges that may be relevant for future studies within the design or coral restoration field.

6.3 Conclusion

This thesis contributes with a new perspective on the role of design in coral reef restoration and how design centred around humans can give new knowledge and aspects into the finished shape of the structure. This shows the importance of developing solutions that can work, not only for the main purpose of the design but also in all the steps of the manufacturing and handling process. To develop a working sustainable solution, it is not enough to only consider the environmental factors. The solution also needs to be practical, economically reasonable possibly to implement in society. Otherwise, it will be difficult to justify its implementation.

7 Future studies

This thesis is based on other studies on what corals and coral larvae prefer. It is therefore essential to evaluate how corals and coral larvae respond to the design. When it comes to the material it is important to understand more about what it is in the volcanic ash that the coral larvae seems to respond positively to, this is based on the study that Förster et al. (2024) did where volcanic ash from the volcano La Soufrière on St. Vincent island was used. Since every volcanic eruption leaves a different combination of metals in the ashes, it is important to understand if material can be collected from every eruption or not.

Design is a reflective process, and it is important to reflect even after the delivery phase of the Double Diamond. The design in this thesis is supposed to make it easier for people working with coral restoration, so the next step would be to have workers try the design and collect user feedback from interviews and observations. After this it is necessary to reflect on what can be improved with the shape in the future. This aligns with the iterative process of human centred design that Norman (2013, s229-230) describes. The transport solution needs further testing as well, since waves and currents could affect safety and handling. A smaller submerged floating device may therefore be a more suitable alternative.

When it comes to the material aspect of the work it needs to be evaluated more thoroughly in a lab. Standardised pressure tests should be done, and the water quality should be tested. According to my research historical examples have shown that lime putty can function as a long-time material, therefore it might still be possible to find a mixture that doesn't contain any Portland cement if more samples, tests and analyses were conducted. It is also crucial to test building a kiln and to collect the material on a remote island to know if this is a realistic solution. To be realistic it needs to be time efficient, cost less money and to make sure there won't be a material shortage.

Future studies should also evaluate how both the material and the shape of the structure perform during longer periods of time in real ocean environments, including exposure to currents, storms and biological growth. The material should also be mixed and tested with armour. I would like to purpose the use of a natural armour, during my interviews I learned that salt water and rebars are not a good mix when it comes to armour. By materials such as coconut fibre or bamboo can sometimes been used as a replacement. This should be analysed and compared to a sample without Armor to ensure what is the best approach.

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Appendix A

Finished schedule

	Monday	Tuesday	Wednesday	Thursday	Friday	Sat/Sun
V. 9 23/2 -1/3	• Introduction • Make a schedule • Look for articles	• Make thesis application • Make diary • Tuition	• Deadline application • Research • Contact users	• Research	• Research • Search exercise	
V. 10 2/3- 8/3	• Research • Deadline search words • Activity analysis	• Research • Email scientist. • Started with the writing	• WS in information search: 13-16	• Going to Italy on student conference • Write thesis	• Italy	• Italy
V. 11 9/3- 15/3	• Italy • Visit the aquarium in Milano for inspiration	• Italy • Write thesis	• Tuitoring • Research • "Affinity mapping"	• Research material • System map • Research different ways to assemble.	• Moodboard • Brainstorm	• Write on thesis
V. 12 16/3 - 22/3	• Tuitoring 13.30 • Brainstorm in group • Research material	• Research how cement is made. • How to make calcium hydroxide of seashells?	• Sketch ideas • CAD early ideas	• PNI • Clay models • 3D print • Brainstorming in group.	• Write thesis	
V. 13 23/3 - 29/3	• Write thesis	• Questions for interview about volcano ash. • Make survey • Analogies.	• Interview with Robin at Ridgehill Minerals • CAD • Send out survey	• Tuition discuss how to make a oven. • Clay models • Brainstorm i group, themes.	• Start with presentation • Activity analysis, find pictures. • Interview with bricklayers. • Make a shopping list for material tests	
V. 14 30/3 -5/4	• Reserch Kiln • Find bricks • Where can the material be bought?	• Write thesis • Shop tools and things for building the kiln	• Pick up clay and bricks • Make a sketch of kiln with measurments • Clay models	• Build the kiln (dry for 48H)	• Fix with presentation. • Video of building the kiln	• Burn seashells.
V.15 6/4- 12/4	• Shop for material tests • Burn more seashells. • Strain burned shells	• Write thesis • Make a function analysis.	• Strain shells • Pick up volcanic ash. • Fix with the presentation	• Make material tests. • Presentation	• Presentation • Make more material tests.	• Analyze the tests. • Presentation
V.16 13/4 -19/ 4	• Deadline presentation: 14.15. • Analyze the tests.	• Fix schedule after feedback from presentation. • Clean after the material tests- • Analyze the tests.	• Clay models • PNI	• Mockups • Evolve the ideas.	• Write thesis.	• SM

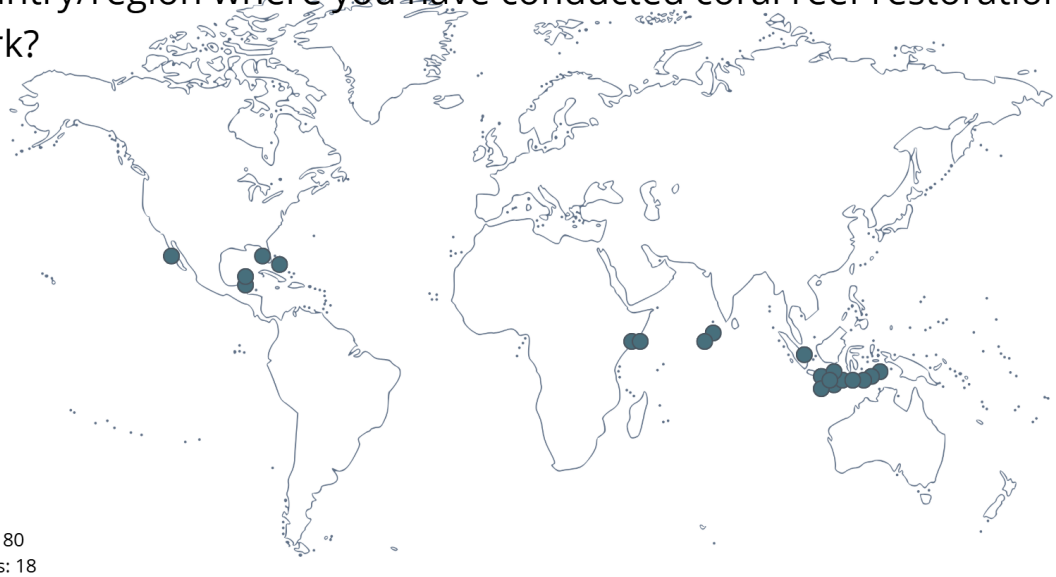
	Mån	Tis	Ons	Tors	Fre	Lör/ Sön
V.17 20/4 - 26/4	• Write thesis • Check the system map. • Evaluate the ideas.	• Develop the ideas • Make sketches • Write thesis	• Write thesis	• Write thesis	• Write thesis	• Going to Serbia
V.18 27/4 -3/5	• Write thesis	• CAD • Tuition	• Start Building model	• CAD • Analyse 3D print and evaluate idea	• Test the material in saltwater. • Schedule. • Write thesis- fix all the appendix.	• Write thesis
V.19 4/5- 10/5	• Build model • Print smal models. • Make changes in CAD?	• Build model • 3D printing	• Build model • 3D printing • Affinity map	• Write thesis- Method part should be done. • Keyshot • New sketches for thesisi, PNI.	• Write thesis- Finish litterature review. • Focus on execution delivery part. • Keyshot, environment picture?	

	Mån	Tis	Ons	Tors	Fre	Lör/ Sön
V.20 11/5 -17/ 5	• Write thesis- delivery part. • Check the rest of the execution part. • CAD mould form and scenario parts. • keyshot	• Plan how everything shoul be visulized and start. • Create enviromental pic. • Thesis- future studies.	• Thesis- result • Continue with CAD mould form. • Render mould parts and transportation.	• Thesis- result • Render- planting and how to assemble. • clean up the function analysis.	• Write theises- conclution and discussion. • Sketch of the finished gestalt.	
V. 21 18/5 -24/ 5	• Deadline thesis • Renders • Enviroment pic. • Zoom-meating 14.45	• Opposition • Poster • Material board	• Presentation • Poster • Paint model • Pictures of how to assemble the function model.	• Opposition • Presentation • Add the last fig. to the thesis	• Opposition • Presentation	• 24/5 Dealine poster
V. 22 25/5 -31/ 5	Final seminar	Final seminar	Final seminar	Build the exhibition?	Opening of exhibition?	
V. 22 1/6- 7/6	• Exhibition?	• Take the exhibition down.				

Appendix B

Survey

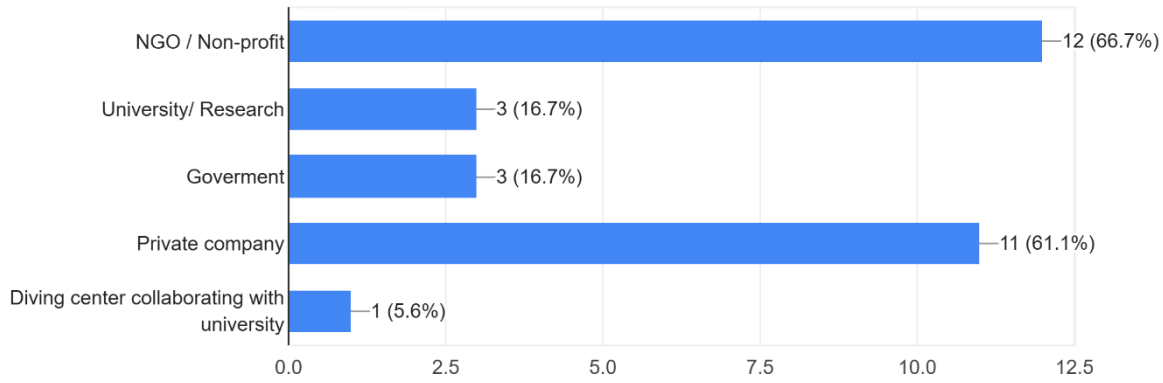
Country/region where you have conducted coral reef restoration work?



Sent to: 80
Answers: 18

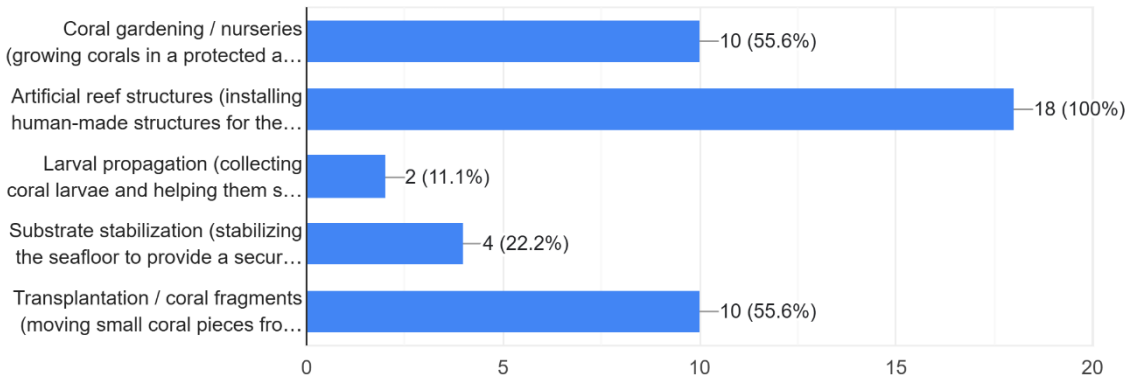
Type of organization you work/ worked with? (Select all that apply)

18 responses



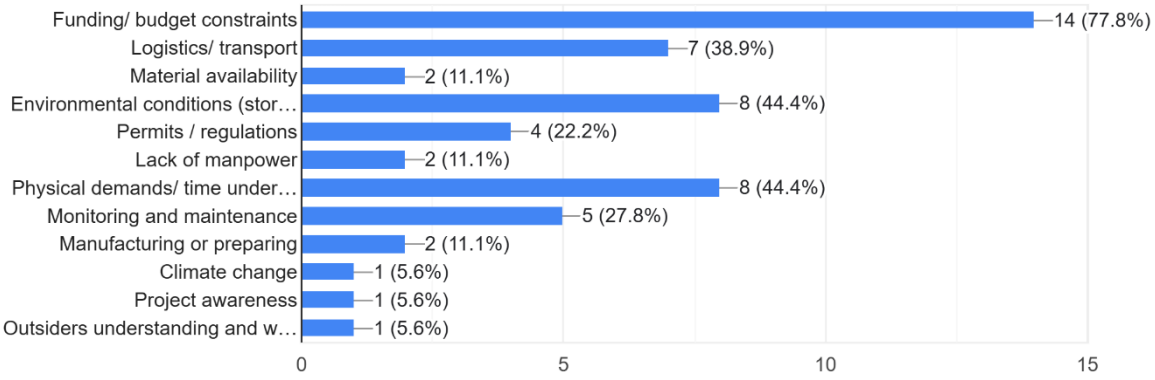
Which coral reef restoration methods have you worked with? (Select all that apply)

18 responses



What are the biggest challenges in your restoration work? (Select up to 3, if you want to leave a comment, choose option "other" as well)

18 responses



Which parts of coral reef restoration work take the most time or effort?

(Answer based on your own experience)

16 responses

Construction of artificial reefs and cleaning of artificial reefs once placed in the water

Moving structures underwater

monitoring and maintenance

The planting of the corals

Maintaining restoration plots

Maintenance of corals before and after outplanting

Simply coral planting. I could spend 3 or 4 hours underwater and I'm efficient. Nevertheless, it takes a long time to do any area. Even with a large crew it can take a long time to make impact at scale

Actually moving the structures was rime consuming but probably what takes the most time is the actual production and bringing the structures into the ocean.

In my opinion was taking the structures to create a bigger and stable area for the fish, we first had to move them and we were not able to move all of the structures that were there because of the amount of time it took us.

Get on sites

Siting correctly under water

Regular monitoring & photographing

Collecting data on coral growth (measuring, photos, keeping track of sample categories)

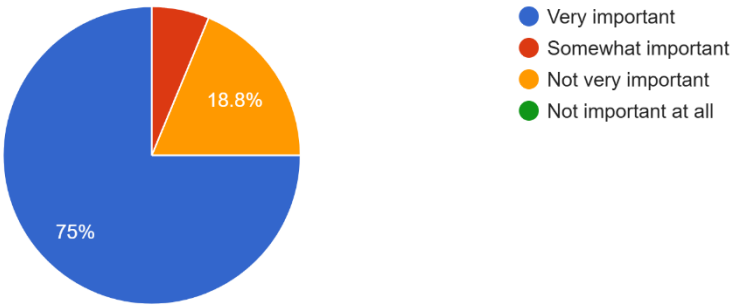
Physically placing the artificial reefs underwater

Installation of artificial reef structures and outplanting

Making and Moving the structures

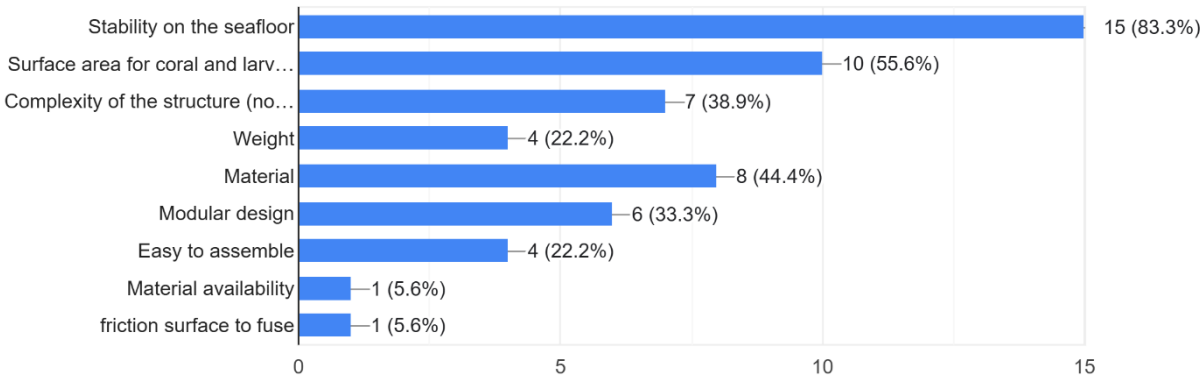
How important is the design/shape of the restoration structures for the success of your work? (if you haven't been working directly with AR structures please answer from you area of knowledge).

16 responses



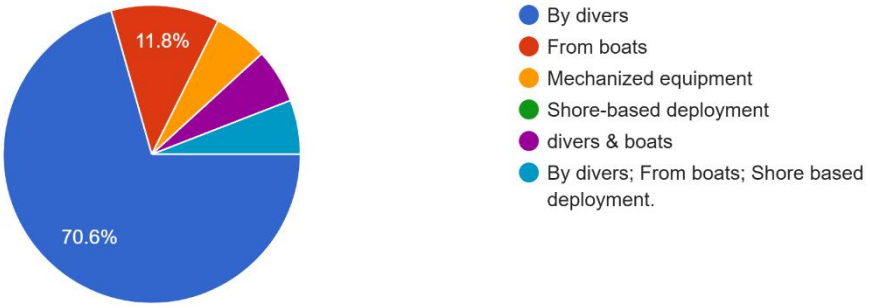
What aspects of structure design are most critical in your experience? (select up to 3, if you want to leave a comment, choose option "other" as well).

18 responses



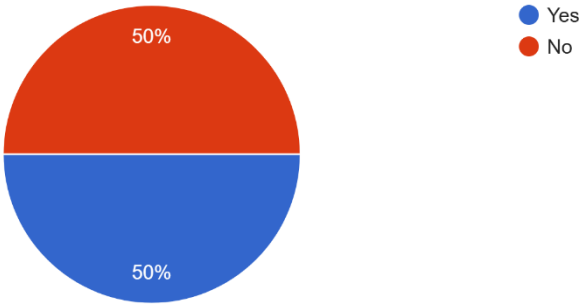
If you have worked with AR structures how were they installed?

17 responses



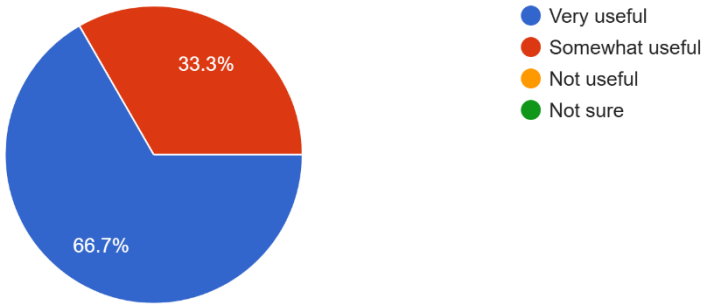
Have you worked with modular or easily assembled structures?

18 responses



If yes, how useful did you find the modularity?

9 responses



Do you believe improved design (for example: shape, structure, material, modularity) of restoration structures could make the work easier or more effective?

15 responses

Yes

We have mostly concentrated on concrete block artificial reefs and coral block artificial reef as these are the most convenient to populate and maintain

Yes, easier for transportation e better for attachment of corals

yes

Yes, but consideration of material in terms of environmental impacts should come before anything else related to design.

Definitely for under-water restoration (sometimes on beach/boat/shall-water). If taking modular, lightweight or easily assembled structures, that's very easy to transport multiple structures at once. Increasing efficiency. However, you'd need to ensure the time saved is greater than the time it then takes to assemble the structure.

Yes, better designs allow the moving of the structure easier!

yeah for sure, it could help the divers and the fishes

Yes, it needs to be resistant to long-term immersion, easy to access locally, and strong enough to stay deep in the substrate to resist extreme weather conditions

definitely, the material used (concrete) was particularly difficult to handle

Yes.

Yes

In your opinion, what could be improved in the design of restoration structures to make work easier, safer, or more effective?

13 responses

Our design allow shore based deployment which makes it cost effective for coastal communities to adopt
easier to install
structures that are easy to clean/brush and to attach coral fragments
A good way to attach them together
Secure attachment to seafloor (to avoid movement during storms)
Most are made from steel rebar. It's super heavy. They're also solid so there's only so many you can take on a boat or install at once. If you could make them foldable or so they can be assembled on the boat, that'd help. If they were lighter, that too. But they'd still need to be durable because underwater things degrade!
The structures that are triangular and not squares were easier to move and place.
Handles to lift structures under water to avoid injuries and be faster easier to move like banchs on the top to be bigger as well, less to carry more efficient
Lighter individual structures that are easy to place but gain strength and weight in situ when connected
Anchorage systems, resilience to rusting
The material or the structures to be lighter and easier to place down, allowing more to be constructed in a shorter amount of time and easier for the divers
Ease of assembling and stability
If they were lighter and easier to move underwater

Appendix C

Questions sent in emails

- What are the challenges you are facing in your work today?
- Would you like to share some pros and cons about the structures you use today or maybe describe a regular day at your base?
- Do you believe that the structures could be further developed to be easier and faster to assemble and place underwater? Is this something that would be beneficial for your work?
- Do you think that a broader variation of shapes could help biodiversity (more species want to move in)?

- Are you using or planning on using more sustainable, locally sourced materials in your structures? If yes, can you tell me more about this?

Appendix D

Interviews

Questions asked to the company Ridgehills minerals:

- How does the process work? For example: Can it be blended with any type of cement?
- Do you know how the ash is collected and whether it is specially processed?
- What are the main differences compared to Portland cement (strength, water resistance, weight, structure of the final product, reduced carbon footprint)?
- Does the volcanic ash require any firing/ pressing afterwards or how does it cure? Do you need any special equipment?
- Are there any critical moments/ difficulties for clients to understand about the product?
- What is the material mainly used for today?
- Has this material been tested in salt water? If yes, what were the results? Did you notice anything different with the water (pH etcetera)?
- Do you know if the material releases any substances/ metals?
- Weight in transport, could it be different compared to transport cement?
- Have you tried mixing the material with anything other than Portland cement?
- Have you observed any biological growth on the material?
- Would it be possible to source the material from different volcanos?
- I have read a little about “Roman concrete” and how they mixed their cement, do you have any knowledge to share about this?

Questions asked to private people working with Portland cement:

- How does the cement process work?
- How do you mix Portland cement?
- How long is the curing time?
- How do you reinforce the material?
- Is a different reinforcement used when producing something that will be in or near salt water?
- Do you have any tips on how to make small samples of different cement mixtures?

Appendix E

List of things needed to build a kiln.

- Bricks (if you have a lot of clay, you don’t need bricks)

- Clay
- Gloves
- Trowel (optional)
- Bucket for water
- Spade
- Measuring tape (I used the bricks to measure with)

Appendix F

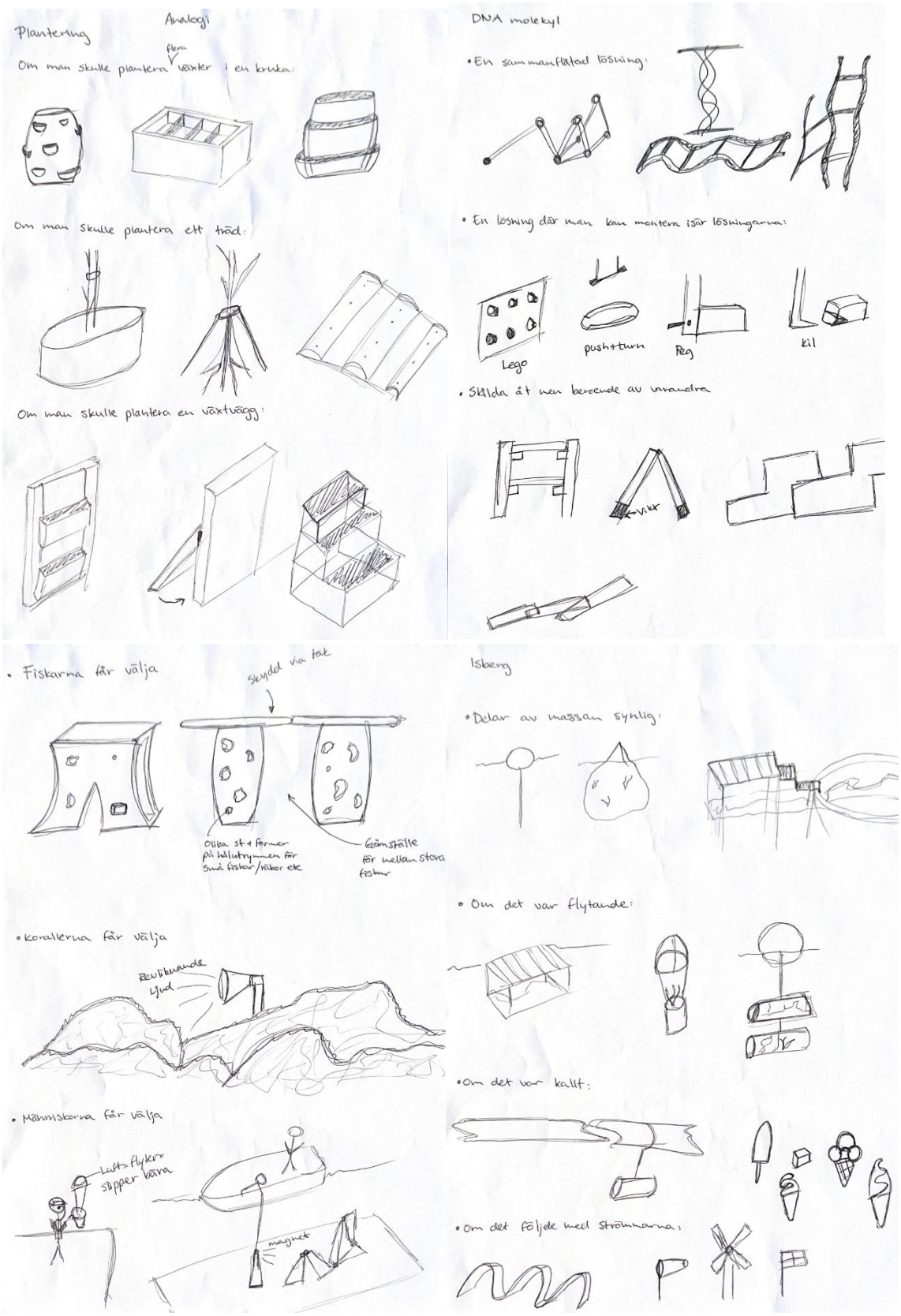
Recipes + analysis material tests

L= Lime putty, C= Cement, A= Ash (volcanic).

Group	Test	Binding	Sand	Water	Result	Comment
Reference	1.0	C 90g A 10g	—	Fresh 17g	Strong, Light colour	Rusty from form. Fizzed with water.
High sand	2.0	C 90g A10g	200g	Fresh 38g	Cracked	Too much sand, was like a sand cake.
Lime base	3.0	L 43g	70g	Fresh 9g	Fraught form, shrank, cracked	Too much water (water in limepuddy).
Lime base	4.0	L 35g	70g	Fresh 6g	Cracked from, pressure, shrank	Less water, needs more binding?
Lime +ash	5.0	L 30g A 20g	50g	Fresh 12g	Cracked from pressure, shrank	Better than 4.0 more binding?
Base 5.0	5.0.1	L 30g A 20g	50g	Salt 12g	Cracked from pressure, shrank	Better than fresh water, fraught.
Base 5.0	5.1	L 30g A 20g	50g	Fresh 8g	Cracked, turned in to powder	Too little water?
Base 5.1	5.1.1	L 30g A 20g	50g	Salt 8g	Powdery surface, cracked.	Too little water? Better than 5.1.
Base 5.0	5.2	L 30g A 20g	50g	Fresh 16g	Sticky surface for 4 days, cracked.	Too little binding?
Base 5.2	5.2.1	L 30g A 20g	50g	Salt 16g	Sticky surface for 4 days, cracked.	Worst of the saltwater ones.
Ash +cement	6.0	C 70g A 30g	—	Fresh 15g	Strong, powdery surface	Fizzed in contact with water.
Hybrid	7.0	L15g/ C40g/ A20g	40g	Fresh 15g	Started to dissolved in water	Fizzed in contact with water.
Base 7.0	7.0.1	L15g/ C40g/ A20g	40g	Salt 15g	Strong, looks best.	Growled in contact with water.
Base 7.0	7.1	L20g/ C30g/ A20g	30g	Fresh 15g	started to dissolved in water	Less sand and cement more lime.
Base 7.0	7.2	L25g/ C20g/ A20g	35g	Fresh 15g	started to dissolved in water	Less cement, more lime.
Base 7.0	7.3	L10g/ C50g/ A20g	20g	Fresh 15g	Strong	Less lime, more cement.
Lime+ ash	8.0	L 50g A 50g	—	Fresh 20g	Fraught form, sticky, cracked	Cracked without pressure, powder.
Lime+ ash	9.0	L 30g A 70g	—	Fresh 30g	Fraught form, sticky, cracked	Cracked with pressure.
Lime+ ash	10.0	L 70g A 30g	—	Fresh 10g	Fraught form, sticky, cracked	Bit stronger than 8 and 9, less water.

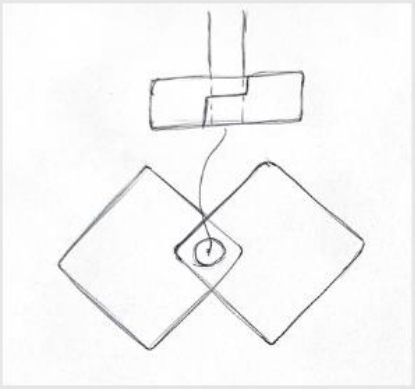
Appendix G

Analogies



Appendix H

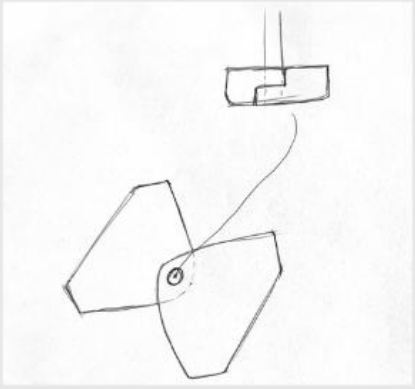
PNI on ideas



Positive:
The shape is easy to manufacture.
Stable.

Negative:
No holes for fishes.
Hard to stabilize in the sand..

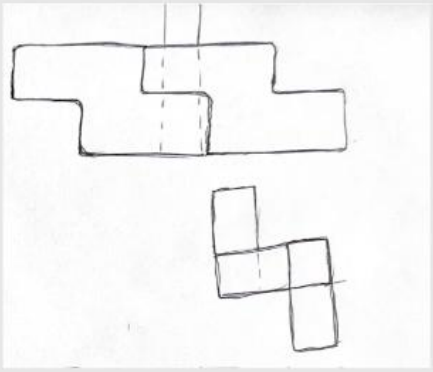
Interesting:
The stick will lock the shapes together.



Positive:
The shape is easy to manufacture.
Can be locked in different ways.

Negative:
No holes for fishes.
Only two figures are
connected to each other.
Hard to stabilize in the sand.

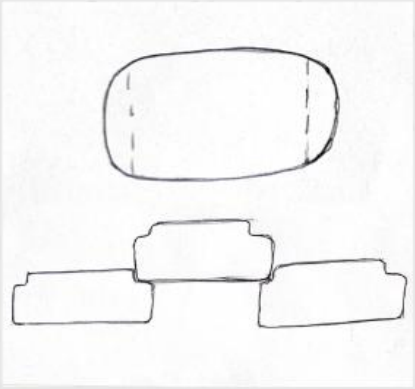
Interesting:
Can be built at height.



Positive:
Can be built in different directions.
Can be built at a slope.

Negative:
Harder to maufacture.
Might need a rebar.
Hard to stabilize in the sand.

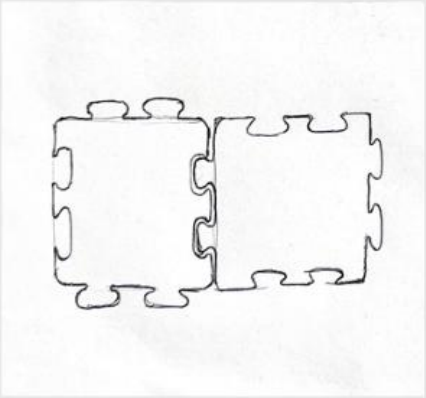
Interesting:
Everything will be connected.



Positive:
No corners , strong shape.
Holes for fishes between the parts.

Negative:
Hard to stabilize in the sand..

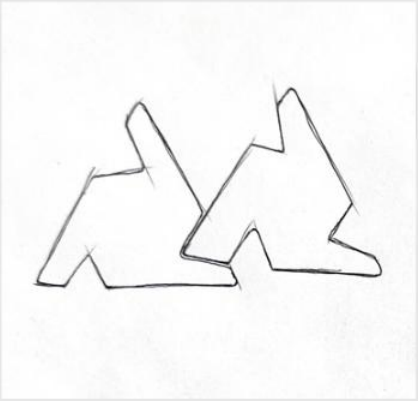
Interesting:
Easy to place.



Positive:
Stable locking.

Negative:
Hard to lock together.
Can only be built in one direction.

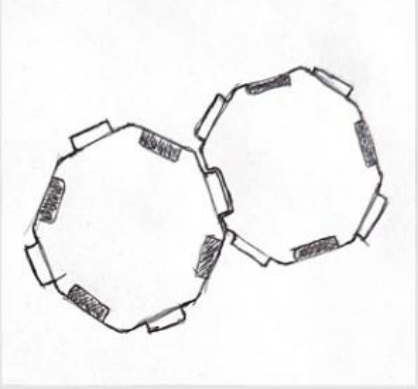
Interesting:
The shape



Positive:
Easy to put together.

Negative:
Doesn't look natural/ organic.
Hard to stabilize in the sand.

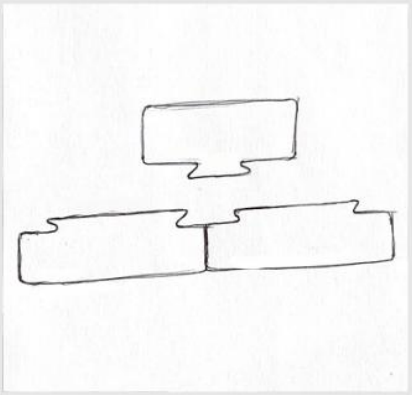
Interesting:
A third shape can be placed above,
that would lock the structures even more.



Positive:
Can be built in different directions.

Negative:
Shape off the locking,
slides apart.

Interesting:
The different levels



Positive:
Stable locking

Negative:
Hard to manufacture.
Fragile parts

Interesting:
Possible to slide



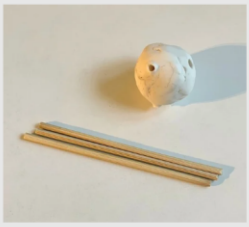
Positive:
Holes in different sizes.
Creates a pattern.

Negative:
Will fall, not stable.

Interesting:
Same base for all the parts.

5 concepts:



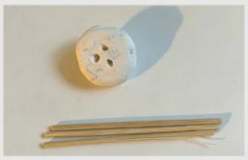


Positive:
Can be placed in any direction.
Stable
"The sticks" lock into each other.

Negative:
The globe might be a bit heavy. (approximately 40 kilos).
The globe will be difficult to transport and carry.
The sticks create a risk, divers can get stuck or rip their gear.
Difficult to assemble, it gets heavier and heavier when you add the sticks, you still need 3 legs so will it be to heavy? (the full structure will approximately have a weight at 85 kilos if the globe needs to be 40 cm above ground).
The sticks will create a load on the globe, this could make the concrete crack.

Interesting:
You can ajust the sticks to get the globe closer or further away from the ground.
It is possible to tie a lace from one stick to another so you can attach corals on the lace.
Possible to have more holes for variation?





Positive:
Looks kind
Natural surface for attaching corals..

Negative:
Looks fragile in the middle. -The legs will create a load from the middle, it will cause the middle concrete part to crack.
Not stable
Will be hard to assemble
Needs longer sticks, more fragile

Interesting:
The shape
The angle off the sticks will make it more or less stable?





Positive:
Most beautiful
Looks like a coral
Will have more similar weight (legs and body).

Negative:
Has a up and a down.
Fragile where the sticks are attached.

Interesting:
Possible to develop the shape.
Possible to have legs in different hights.



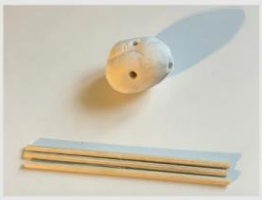


Positive:
Very stable
Can stand on a angled surface
The sticks lock into each other.
The globe is small.

Negative:
Fragile globe
Where to attach corals?
Creating a risk for the divers? Can get stuck on the stick?

Interesting:
Possible to get a variation on the space underneath depending on where you place the globe on the stick.
Attach corals on a lace hanging from the stick?





Positive:
The sticks lock into each other.
Can be placed in different ways.

Negative:
Will roll away in currents.
The globe looks heavy.
The globe will be hard to transport.
The legs will create a load on the globe.
The sticks create a risk, divers can get stuck or rip their gear.

Interesting:
It can be placed both with the globe down and the stick down.